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(54) **IMAGE FORMING APPARATUS AND CARTRIDGE**

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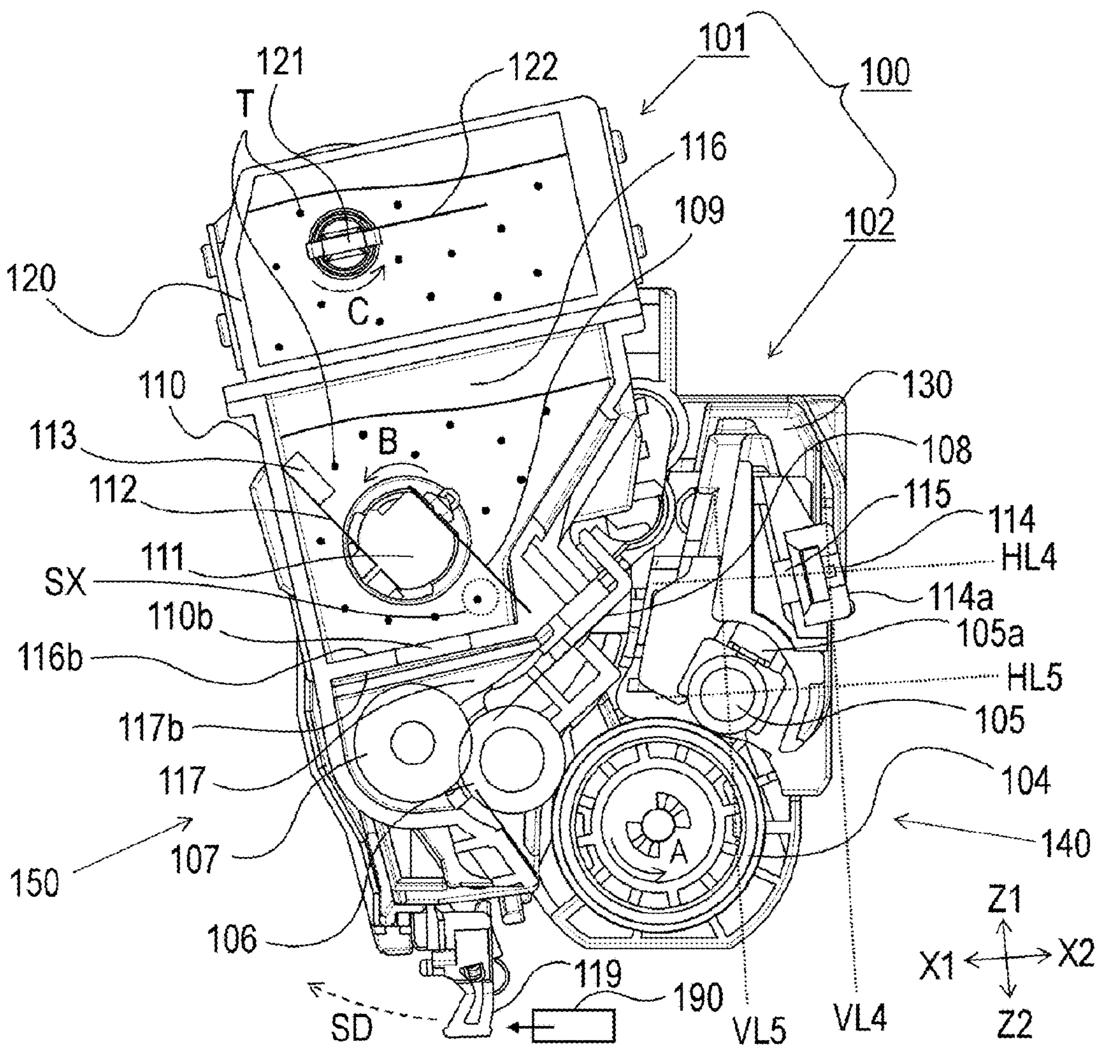
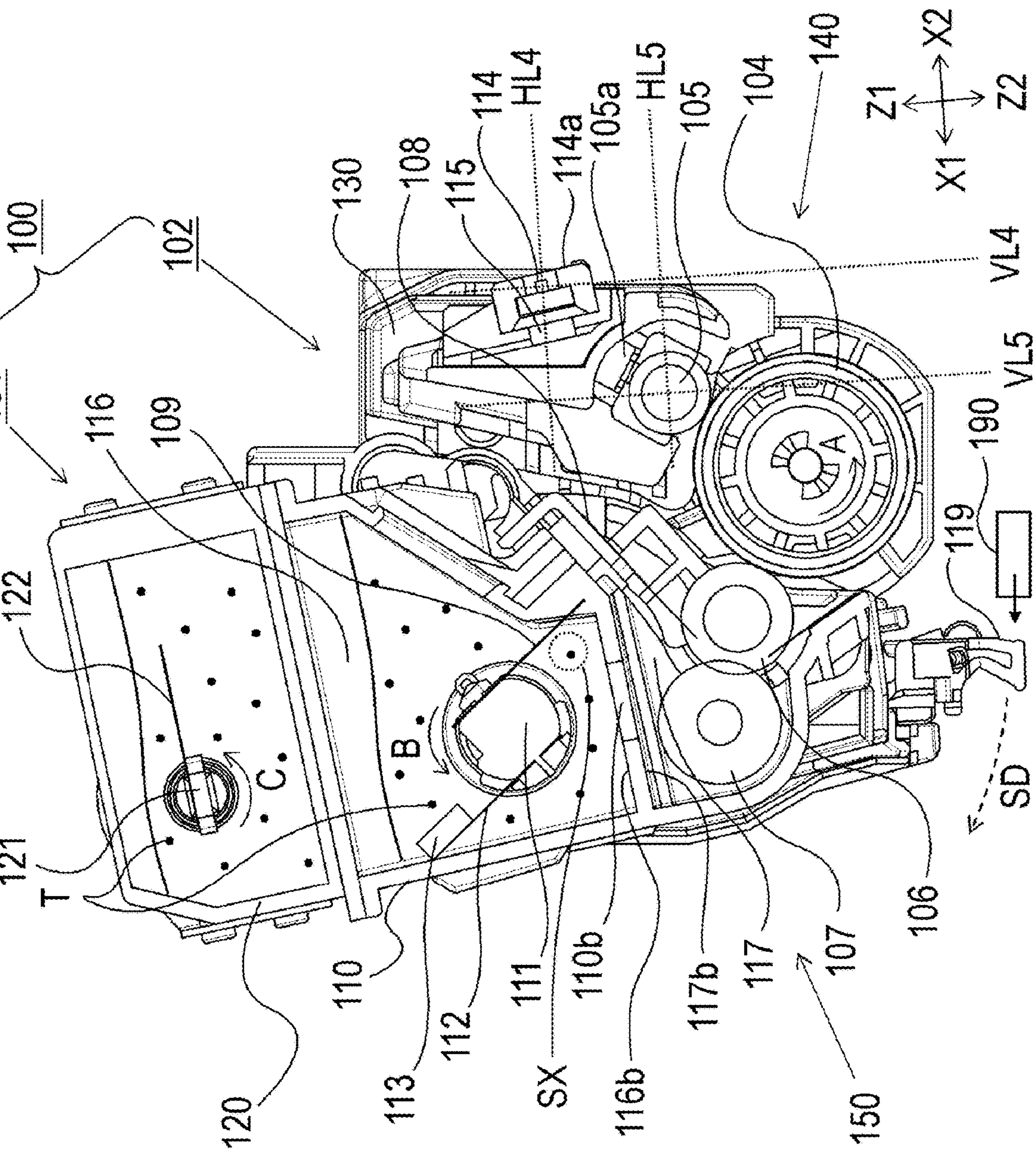
(57) **ABSTRACT**

In an image forming apparatus including an apparatus main body, a first cartridge that is detachably attachable to the apparatus main body, and a second cartridge that is detachably attachable to the apparatus main body, the first cartridge includes a storage chamber that stores developer, a rotational shaft that is provided in the storage chamber so as to be rotatable about a rotational axis line, a flexible member that has one end and an other end in a direction crossing the rotational axis line, the one end being a free end and the other end being a fixed end fixed to the rotational shaft, and a magnetism generating member that is fixed between the fixed end exclusive and the free end inclusive of the flexible member, and the second cartridge includes a sensor for detecting magnetism generated by the magnetism generating member.





100
101
102
130
108
115
114
HL4
114a
105a
HL5
105
104
140
Z1
X2
X1
Z2
VL4
VL5
A
190
119
SD
106
107
117
117b
116b
110b
SX
111
112
113
110
120
T
121
122
116
109
B
C
150

FIG. 1

FIG. 2

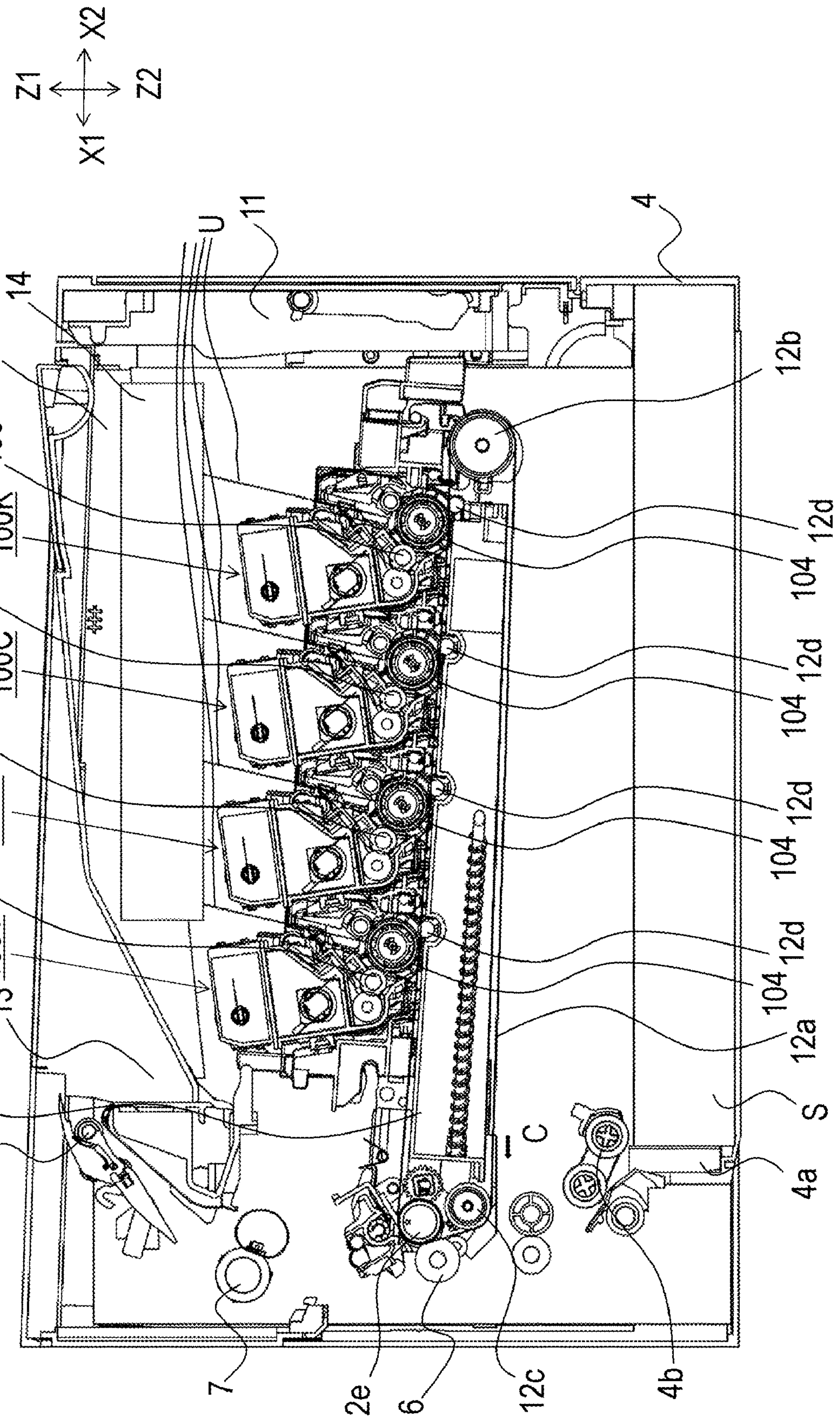

X2
Z1
Z2
X1
M
U
11
4
14
170
12b
106
100K
12d
104
106
100C
12d
104
106
100M
12d
104
106
100Y
12d
104
13
12a
12
S
C
8
4a
7
12e
6
12c
4b

FIG. 3

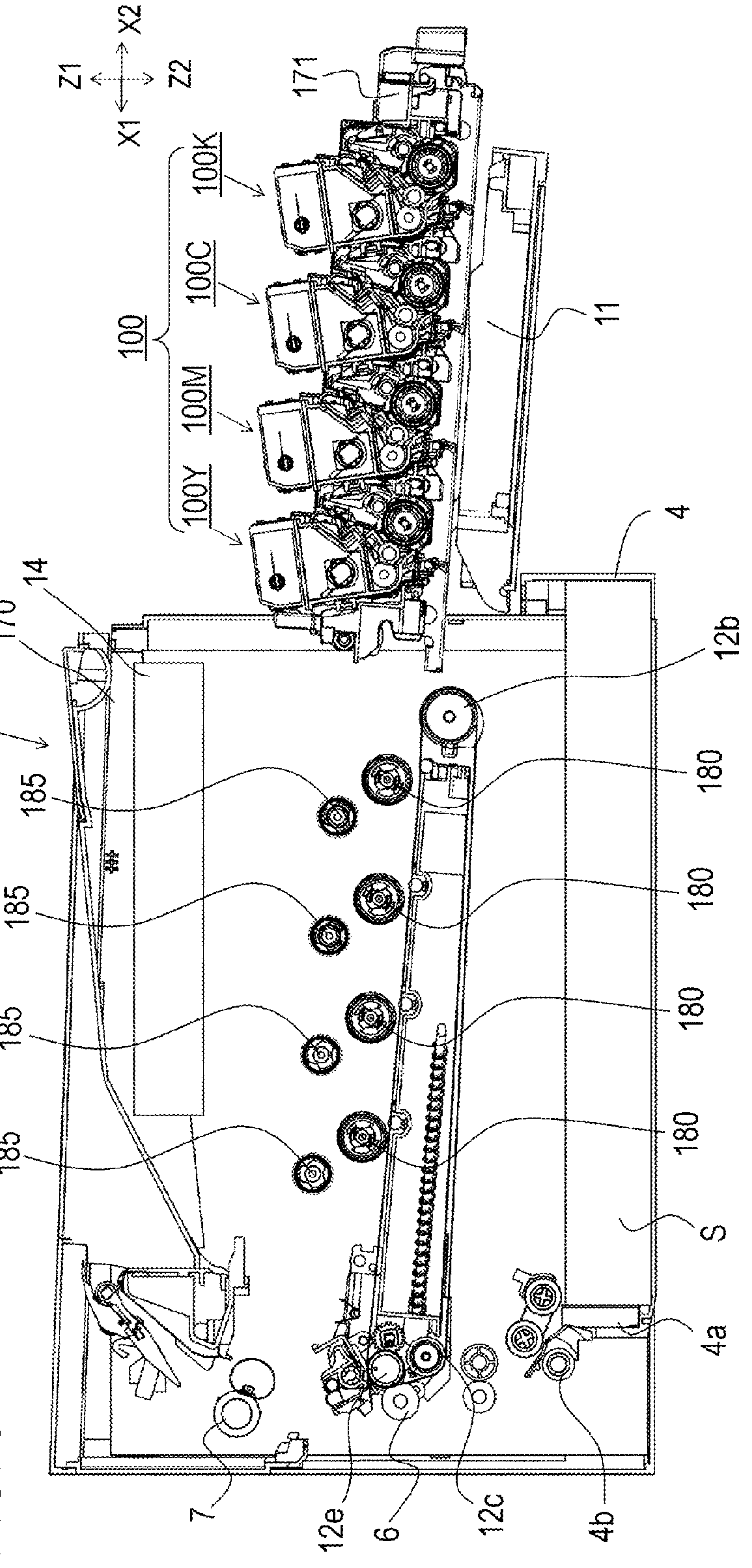
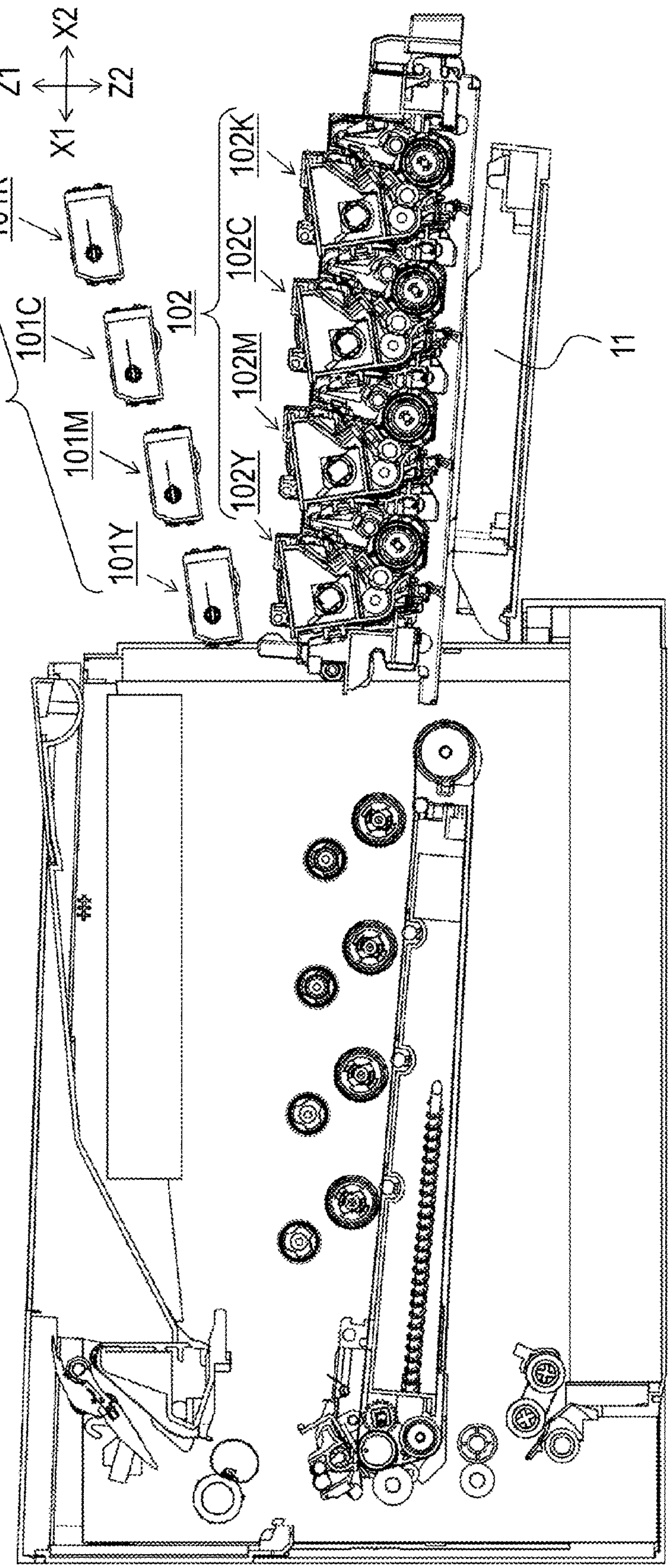

X1
X2
Z1
Z2
171
100K
100C
100M
100Y
100
11
4
14
170
M
12b
180
180
180
180
185
185
185
185
S
4a
4b
12c
6
12e
7

FIG. 4



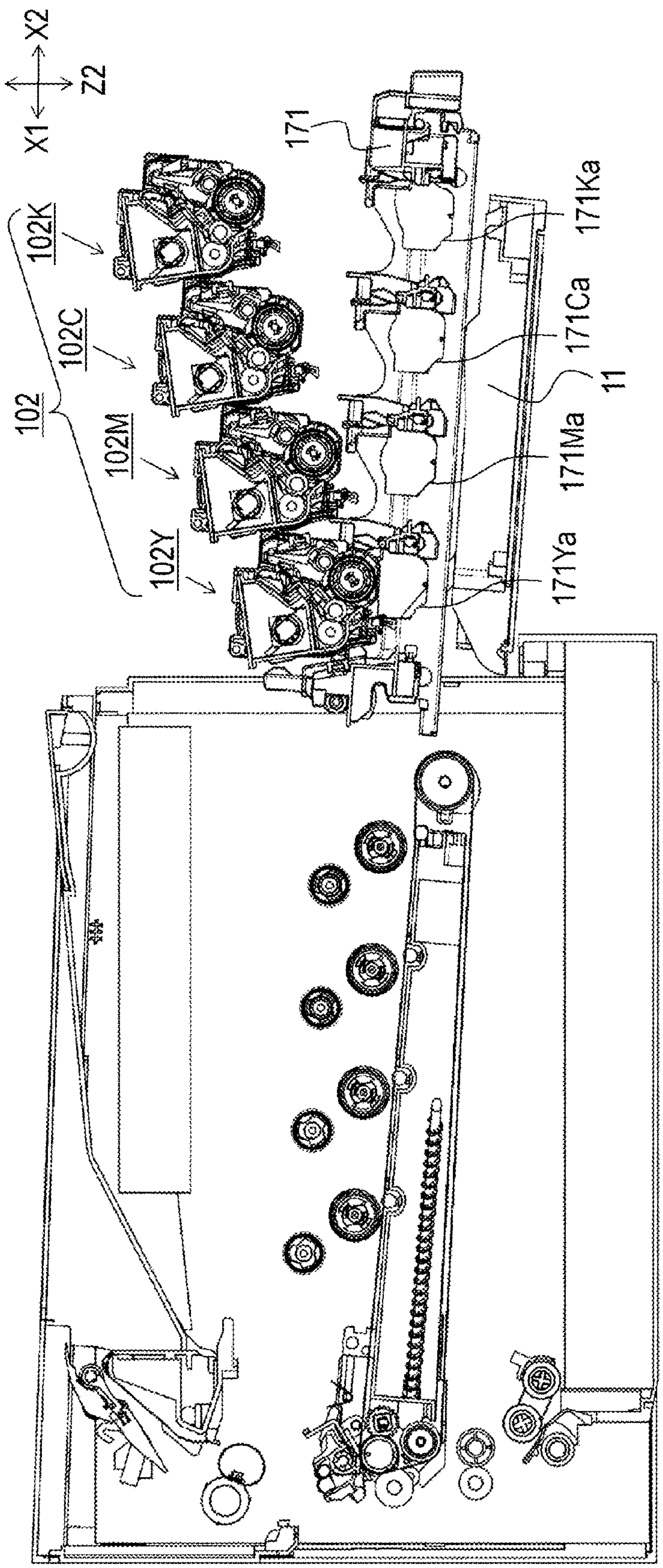

X1
X2
Z1
Z2
102
102K
102C
102M
102Y
171
171Ka
171Ca
171Ma
171Ya
11

FIG. 5

FIG. 6B

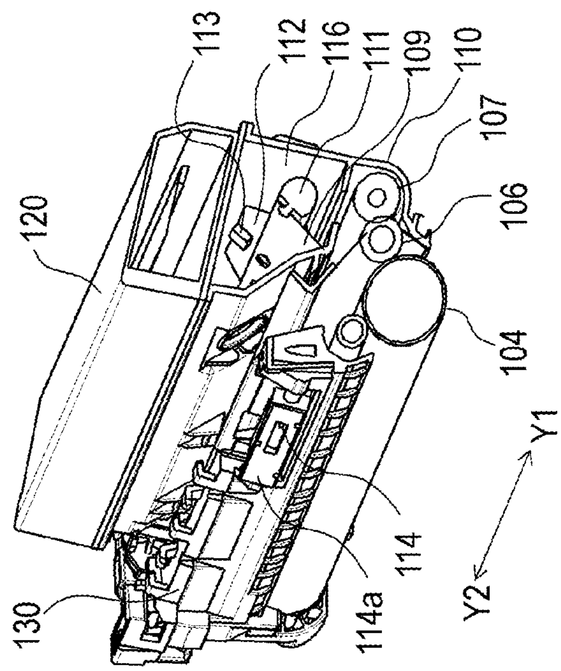

113
112
116
111
109
110
107
106
104
120
114
114a
130
Y1
Y2

FIG. 6A

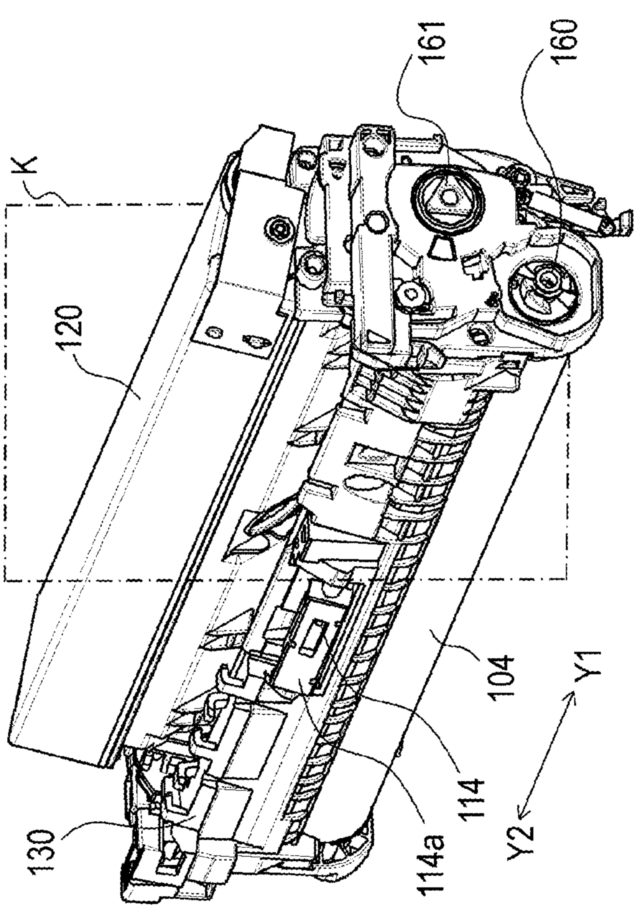

K
120
161
160
104
114
114a
130
Y1
Y2

FIG. 7

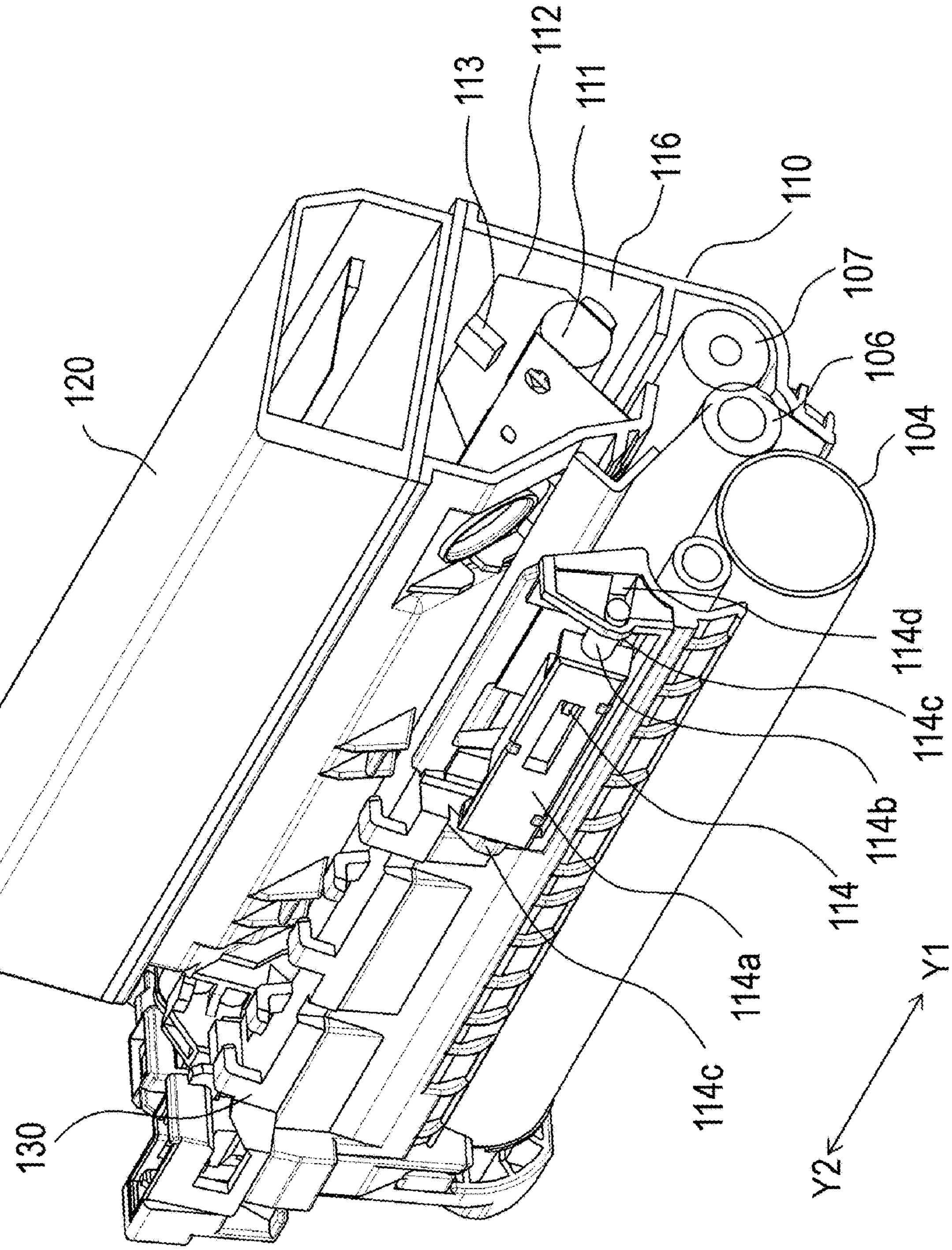

120
113
112
111
116
110
107
106
104
114d
114c
114b
114
114a
114c
130
Y1
Y2

FIG. 8

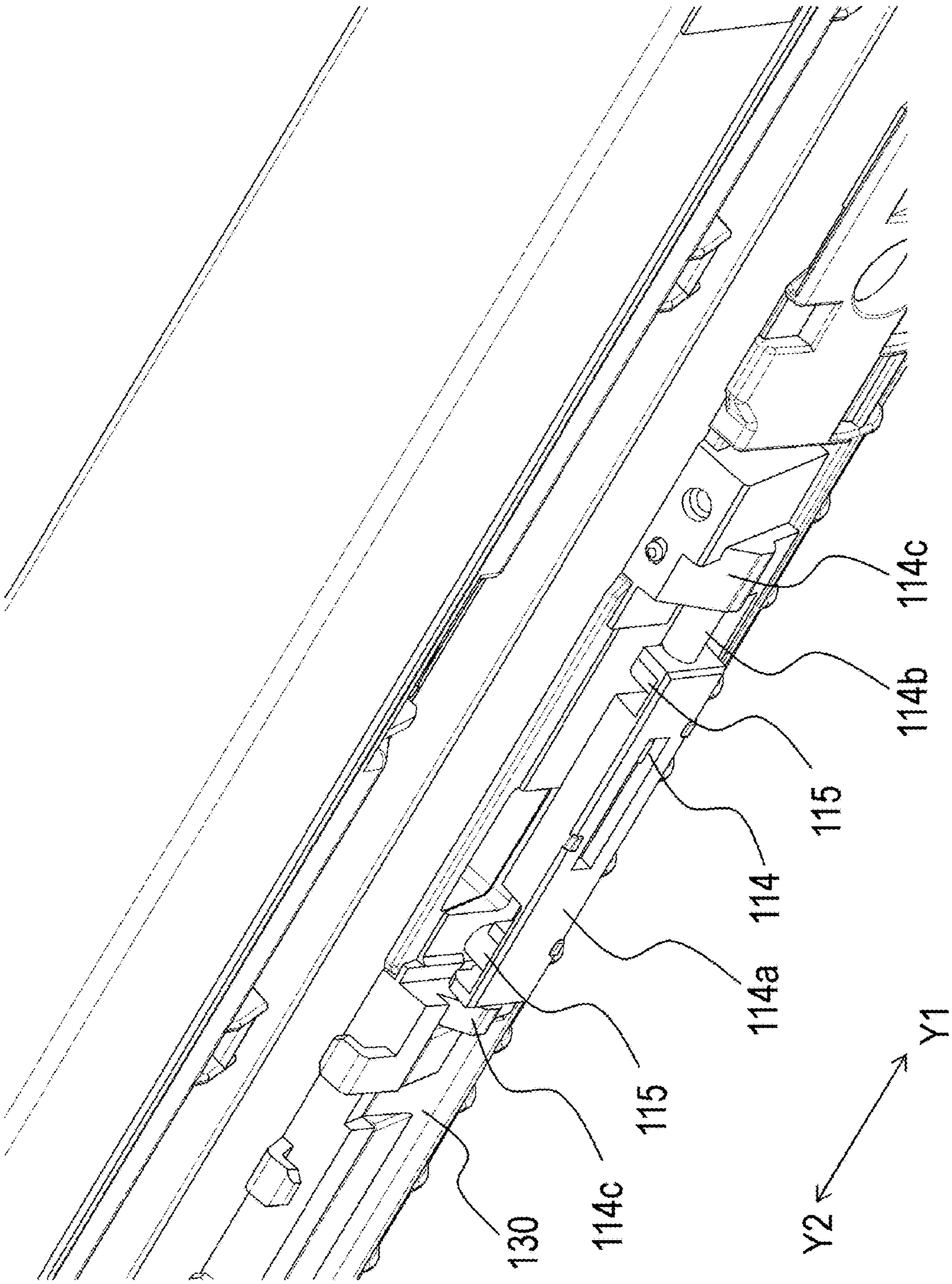
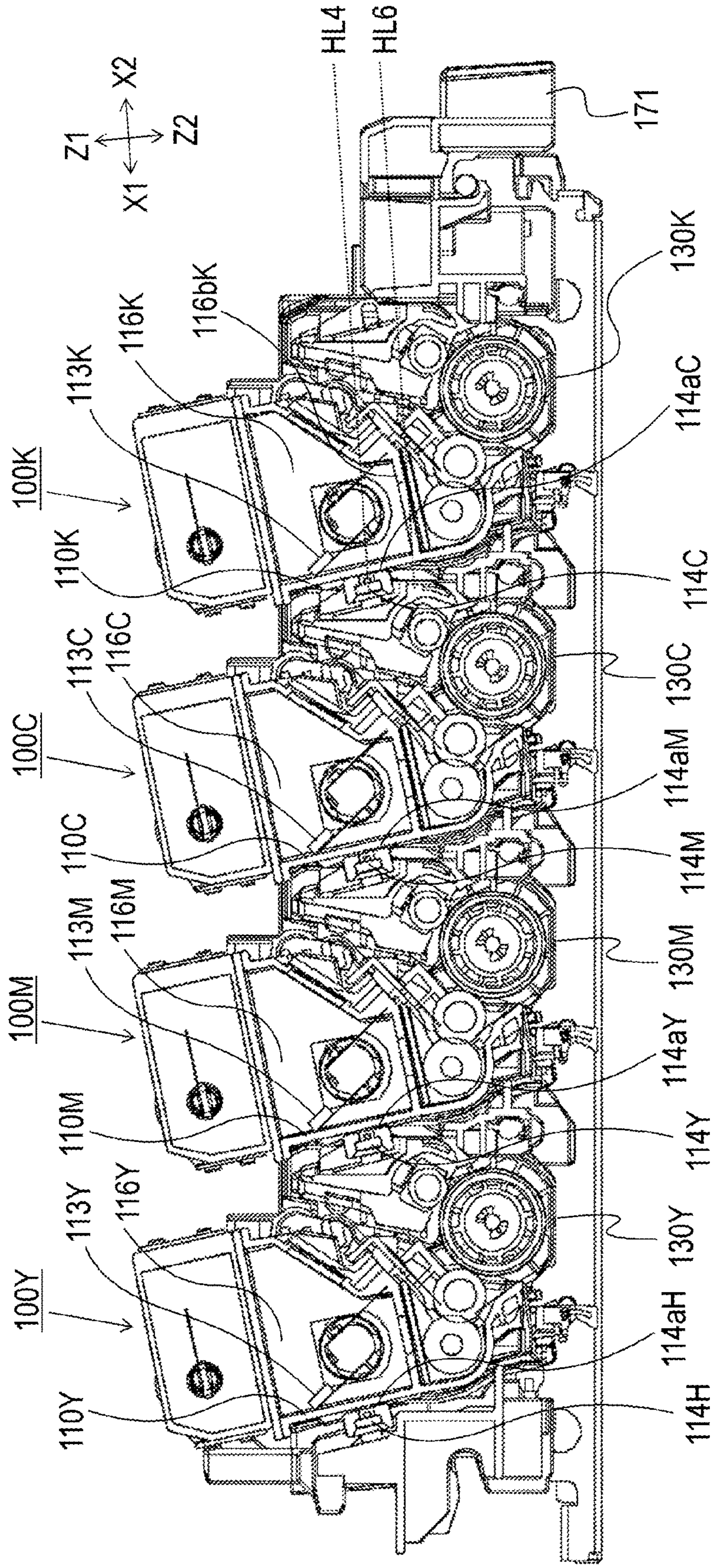


FIG. 9



X1
X2
Z1
Z2
HL4
HL6
171
100K
100C
100M
100Y
113K
116K
116bK
110K
130K
114aC
114C
113C
116C
110C
130C
114aM
114M
113M
116M
110M
130M
114aY
114Y
113Y
116Y
110Y
130Y
114aH
114H

FIG. 10A

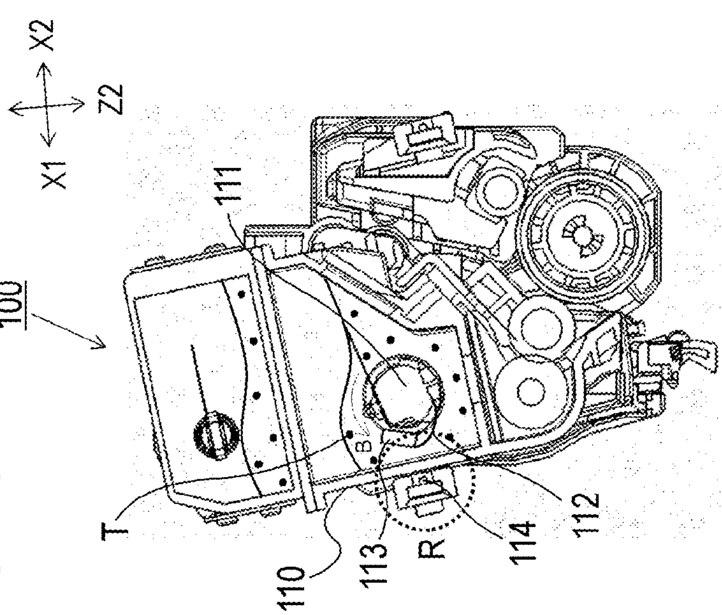


FIG. 10B

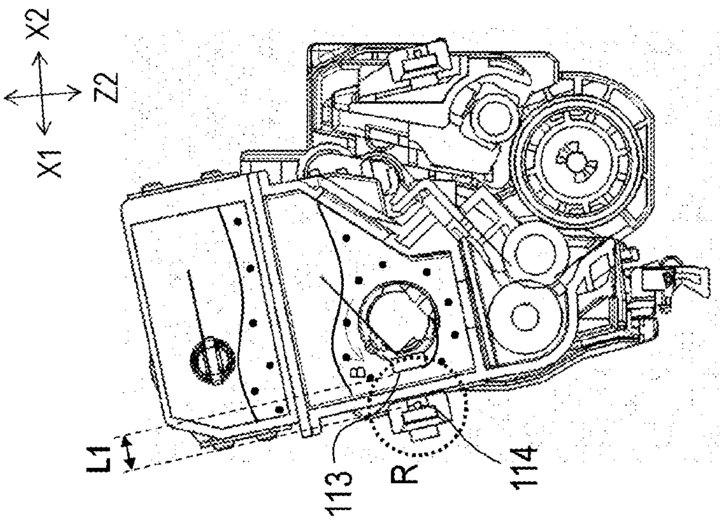


FIG. 10C

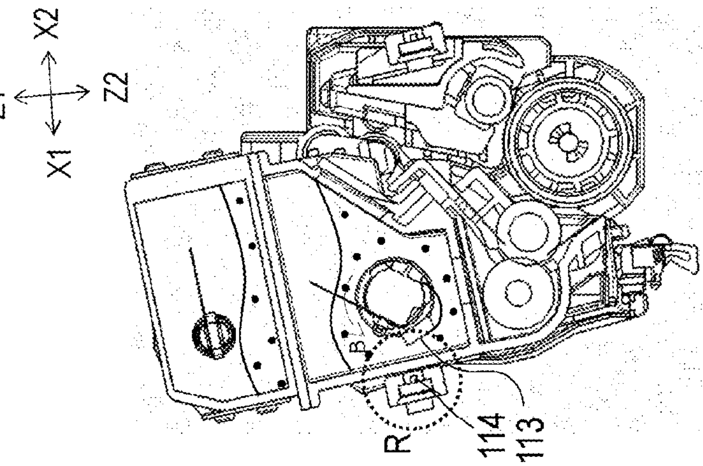


FIG. 11C

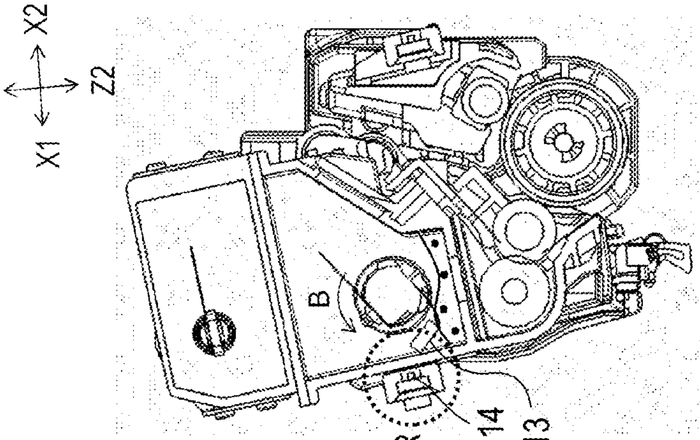


FIG. 11B

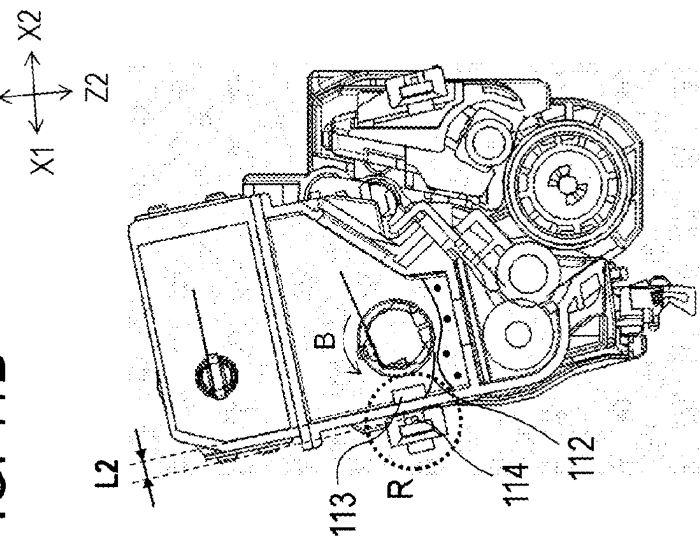


FIG. 11A

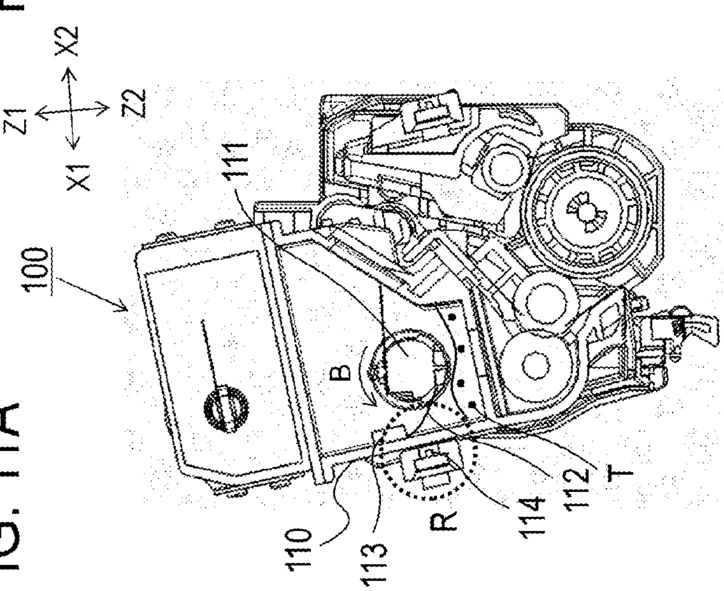


FIG. 12A

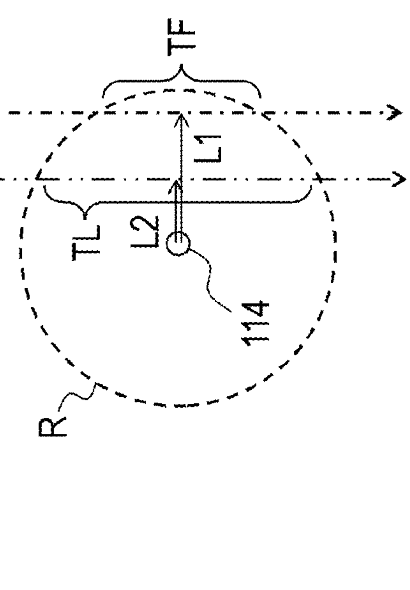


FIG. 12B

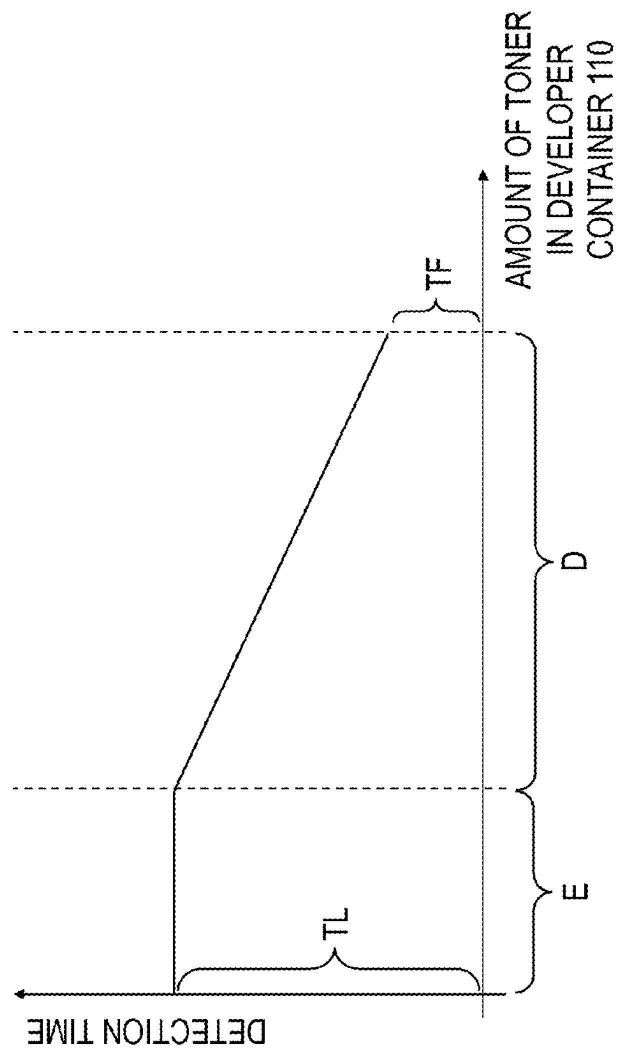


FIG. 13

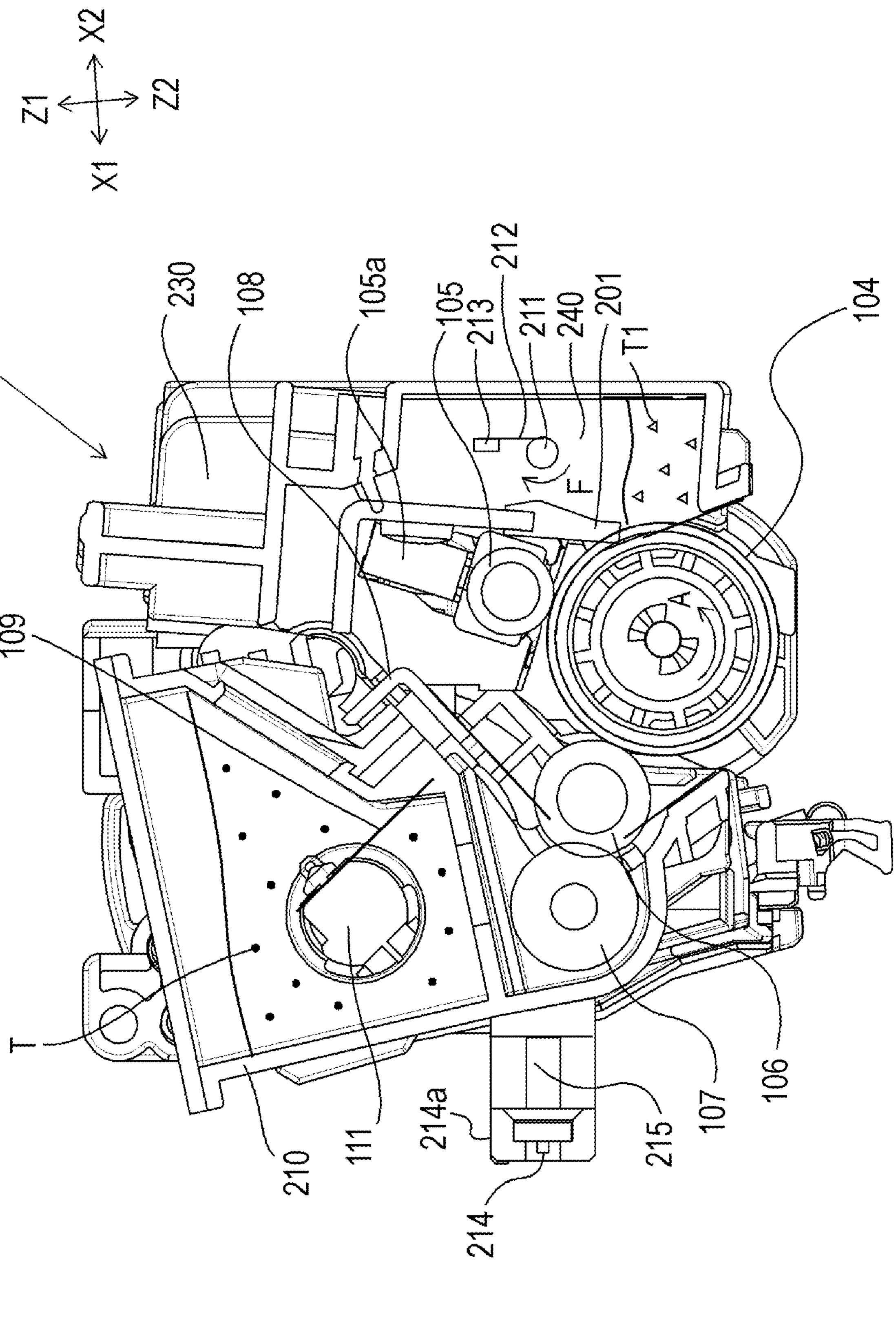

X2
X1
Z1
Z2
200
230
108
105a
105
213
212
211
240
201
T1
F
104
A
109
T
210
111
214a
214
215
107
106

FIG. 14

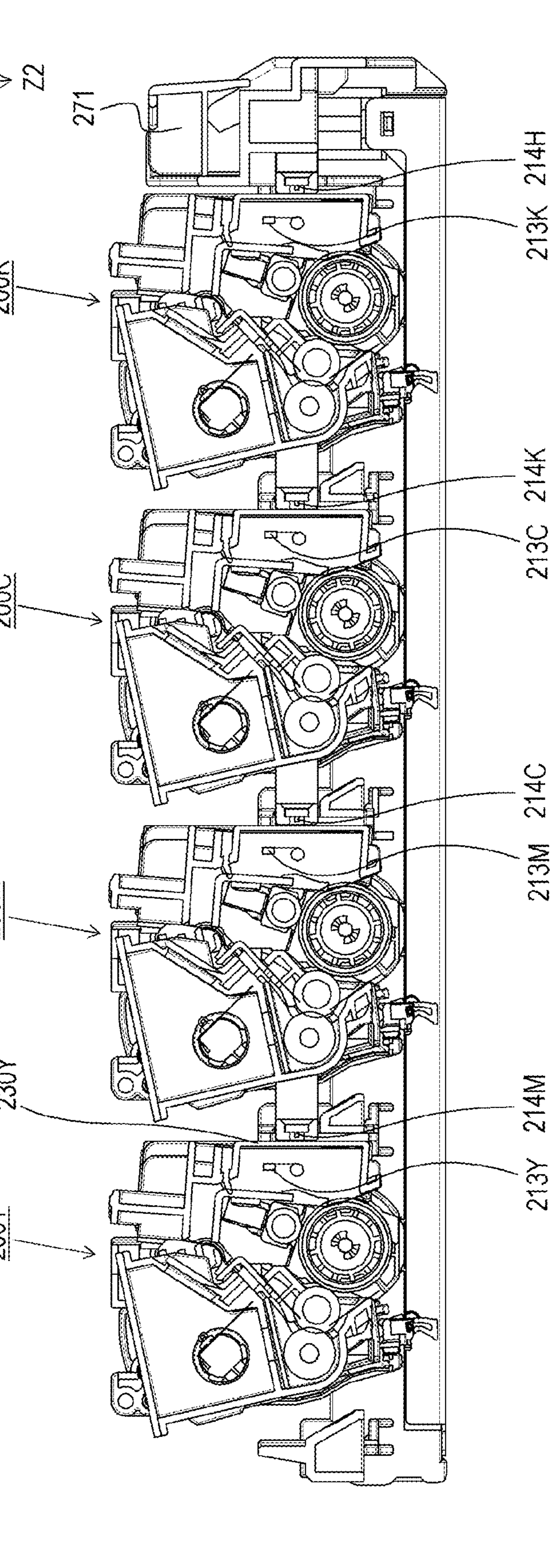

X2
Z1
Z2
X1
271
214H
213K
200K
214K
213C
200C
214C
213M
200M
230Y
214M
213Y
200Y

FIG. 15A

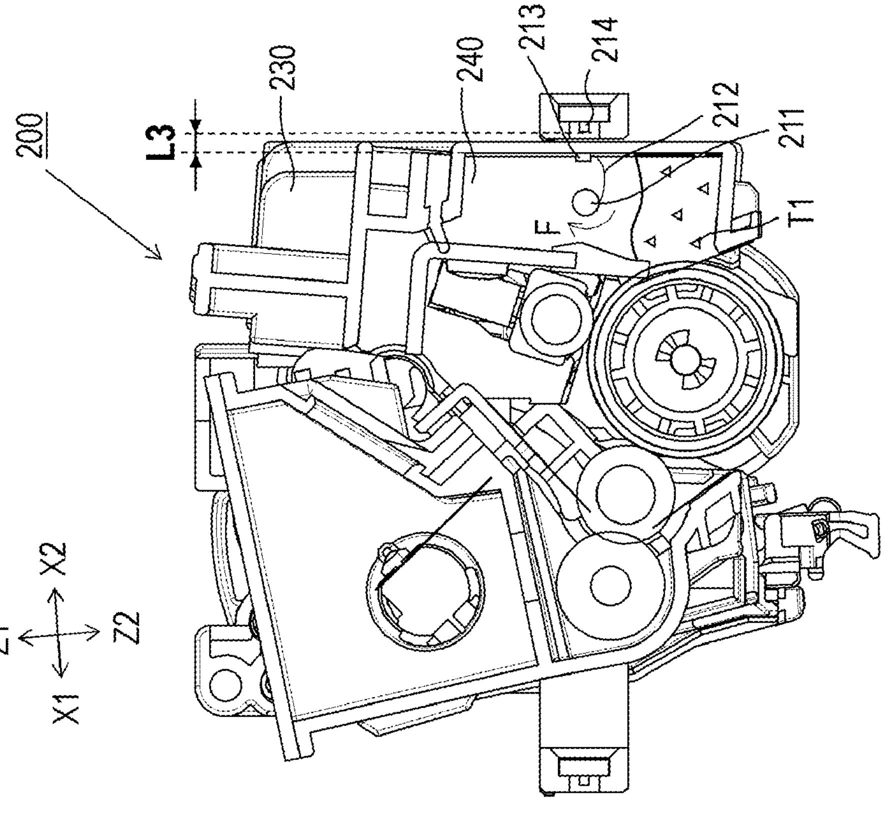


FIG. 15B

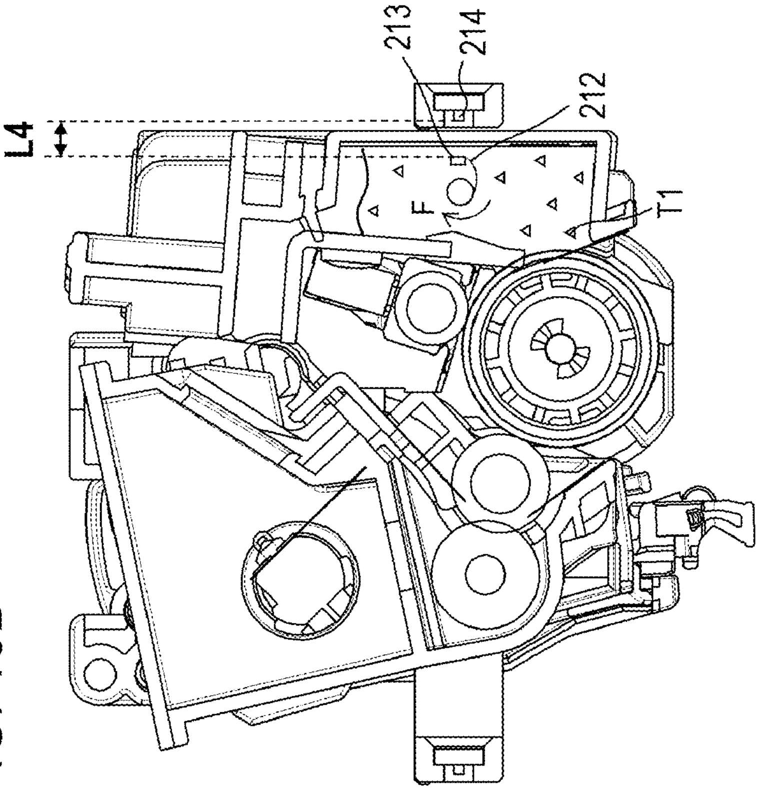


FIG. 16

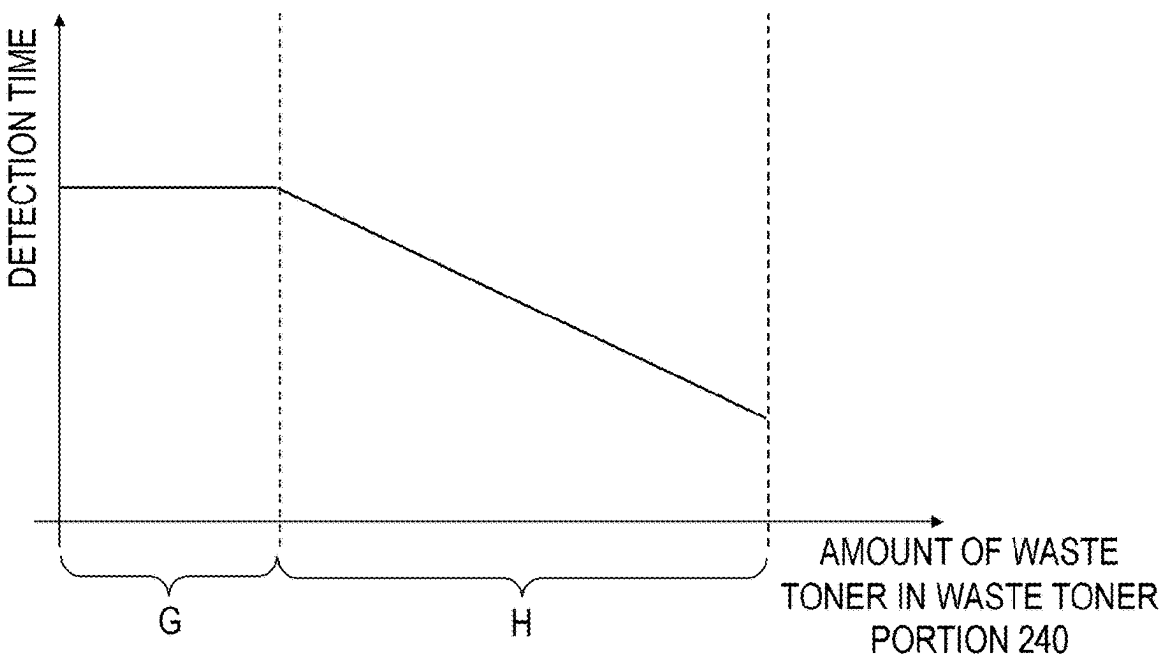

DETECTION TIME
AMOUNT OF WASTE TONER IN WASTE TONER PORTION 240
G
H

IMAGE FORMING APPARATUS AND CARTRIDGE

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to a cartridge detachably attachable to an image forming apparatus employing an electrophotographic method.

Description of the Related Art

[0002] In an electrophotographic image forming apparatus employing a so-called cartridge system, various configurations are employed as a configuration for detecting a remaining amount of toner as developer and informing a user of timing to exchange the cartridge. While an optical remaining-toner-detection method and an electrostatic remaining-toner-detection method are typical hardware methods for detecting the remaining amount of toner, a remaining-toner-detection method using a magnet is also proposed in Japanese Patent Application Publication No. 2004-286849 and Japanese Examined Utility Model Application Publication No. H01-032049. In the remaining-toner-detection method using a magnet, when the position of the magnet installed inside a container changes in accordance with the amount of toner therein, the distance between the magnet and a sensor changes, and based on a detection voltage changed due to this change in the distance, the amount of toner is estimated. The optical remaining-toner-detection method needs a light guide for input and output, and thus, a space is needed for the light guide. In the electrostatic remaining-toner-detection method, since a high voltage is applied for input and output, an insulation distance needs to be provided to prevent leakage of electricity, and thus, a space is also needed. In contrast, in the remaining-toner-detection method using the magnet as disclosed in Japanese Patent Application Publication No. 2004-286849 and Japanese Examined Utility Model Application Publication No. H01-032049, since the magnet installed inside the container corresponds to the input, there is no need for providing a space for the light guide or the insulation distance.

SUMMARY OF THE INVENTION

[0003] In the market of full-color image forming apparatuses in recent years, miniaturization is essential for product competitiveness, and there is a demand for a more compact configuration in the system for detecting the remaining amount of toner.

[0004] An object of the present invention is to provide a configuration for detecting the remaining amount of developer, the configuration facilitating apparatus miniaturization.

[0005] In order to solve the above-described problem, an image forming apparatus according to the present invention includes:

- **[0006]** an apparatus main body;
- **[0007]** a first cartridge that is detachably attachable to the apparatus main body, the first cartridge including a storage chamber that stores developer, a rotational shaft that is provided in the storage chamber so as to be rotatable about a rotational axis line, a flexible member that has one end and an other end in a direction crossing the rotational axis line, the one end being a free end and the other end being a fixed end fixed to the rotational shaft, and a magnetism generating member that is fixed between the fixed end exclusive and the free end inclusive of the flexible member; and
- **[0008]** a second cartridge that is detachably attachable to the apparatus main body, the second cartridge including a sensor for detecting magnetism generated by the magnetism generating member.

[0009] In order to solve the above-described problem, a cartridge according to the present invention includes:

- **[0010]** a storage chamber that stores developer;
- **[0011]** a rotational shaft that is provided in the storage chamber so as to be rotatable about a rotational axis line;
- **[0012]** a flexible member that has one end and an other end in a direction crossing the rotational axis line, the one end being a free end and the other end being a fixed end fixed to the rotational shaft; and
- **[0013]** a magnetism generating member that is fixed between the fixed end exclusive and the free end inclusive of the flexible member,
- **[0014]** wherein the cartridge is one of a plurality of cartridges detachably attachable to an apparatus main body of an image forming apparatus, and
- **[0015]** wherein the cartridge includes a sensor for detecting magnetism generated by an other magnetism generating member included in an other cartridge among the plurality of cartridges attached to the apparatus main body.

[0016] According to the present invention, there can be provided a configuration for detecting the remaining amount of developer, the configuration facilitating apparatus miniaturization.

[0017] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. **1** is a sectional view of a cartridge according to Embodiment 1;

[0019] FIG. **2** is a sectional view of an image forming apparatus according to Embodiment 1;

[0020] FIG. **3** is a sectional view of the image forming apparatus according to Embodiment 1;

[0021] FIG. **4** is a sectional view of the image forming apparatus according to Embodiment 1;

[0022] FIG. **5** is a sectional view of the image forming apparatus according to Embodiment 1;

[0023] FIGS. **6**A and **6**B are a perspective view and a sectional perspective view of the cartridge according to Embodiment 1;

[0024] FIG. **7** is a sectional perspective view of the cartridge according to Embodiment 1;

[0025] FIG. **8** is a perspective view of the periphery of a sensor of the cartridge according to Embodiment 1;

[0026] FIG. **9** is a sectional view of a tray unit according to Embodiment 1;

[0027] FIGS. **10**A to **10**C are explanatory diagrams illustrating a method for detecting the remaining amount of toner according to Embodiment 1;

[0028] FIGS. **11**A to **11**C are explanatory diagrams illustrating the method for detecting the remaining amount of toner according to Embodiment 1;

[0029] FIGS. **12**A and **12**B are explanatory diagrams illustrating a relationship between detection time and the remaining amount of toner according to Embodiment 1;

[0030] FIG. **13** is a sectional view of a cartridge according to Embodiment 2;

[0031] FIG. **14** is a sectional view of a tray unit according to Embodiment 2;

[0032] FIGS. **15**A and **15**B are explanatory diagrams illustrating a method for detecting the remaining amount of toner according to Embodiment 2; and

[0033] FIG. **16** is an explanatory diagram illustrating a relationship between detection time and the amount of waste toner according to Embodiment 2

DESCRIPTION OF THE EMBODIMENTS

[0034] Hereinafter, embodiments of the present invention will be described in detail with reference to the drawings. However, the dimensions, materials, shapes, relative arrangements, and the like of the components described in the embodiments are to be appropriately changed according to a configuration and various conditions of the apparatus to which the invention is applied. That is, the scope of the present invention is not limited to the following embodiments. Further, although a plurality of features is described in the embodiments, all of the plurality of features are not necessarily essential to the invention, and the plurality of features may be arbitrarily combined. In the accompanying drawings, the same or similar components are denoted by the same reference numerals, and redundant description will be omitted.

Embodiment 1

[0035] An electrophotographic image forming apparatus (hereinafter, also referred to as an "image forming apparatus") according to Embodiment 1 of the present invention will be described with reference to the drawings. The electrophotographic image forming apparatus forms an image on a sheet-like recording medium such as paper by using an electrophotographic image forming method. Specific examples of the image forming apparatus include a copier, a facsimile machine, a printer (a laser beam printer, an LED printer, etc.), and a multifunction peripheral (a multifunction printer) of the above apparatuses. In the embodiment described below, a laser beam printer will be described as an example as one aspect of the image forming apparatus.

[0036] In Embodiment 1, an image forming apparatus to which four toner cartridges and four process cartridges are detachably attachable as cartridges will be described as an example of the image forming apparatus. The number of toner cartridges and the number of process cartridges attached on the image forming apparatus are not limited to these numbers, and are appropriately set as needed.

[0037] The cartridges are units which are fixed to or detachably attachable to the above-described image forming apparatus (a detachably attachable unit is also referred to as a cartridge). Examples of the unit configuration include a unit having process member (for example, a charging member, a developing member, a cleaning member, etc.) acting on a photosensitive member, a unit having toner, etc. That is, the cartridge could have various modes. For example, the cartridge could have a mode in which an all-in-one cartridge in which a drum unit and a developing unit are integrated is replaced, a mode in which a drum unit and a developing unit are separately replaced, etc. Further, a mode in which a process cartridge, other than a toner container containing toner, is fixed to the image forming apparatus, and when the toner is used up, the toner container is replaced by a user, or a mode in which both the toner container and the process cartridge are replaced may also be used.

Schematic Configuration of Image Forming Apparatus

[0038] FIG. **1** is a sectional view of a cartridge **100** according to Embodiment 1 of the present invention. FIG. **2** is a sectional view schematically illustrating a configuration of an image forming apparatus M according to Embodiment 1 of the present invention.

[0039] The image forming apparatus M is a full-color laser printer using a 4-color electrophotographic process, and forms a color image on a recording medium S. The image forming apparatus M employs a cartridge system, and forms a color image on a recording medium S by detachably attaching the cartridge **100** on an image forming apparatus main body **170**.

[0040] A side on which a front door **11** is provided is a front side (a front surface) of the image forming apparatus M, and a side opposite to the front side is a rear side (a rear surface) of the image forming apparatus M. Further, the right side and the left side of the image forming apparatus M as viewed from the front side are referred to as a drive side and a non-drive side, respectively.

[0041] FIGS. **2** to **5** illustrate states in each of which the image forming apparatus M is placed on a horizontal installation surface, which is a normally assumed installation state of the image forming apparatus M. In each of FIGS. **2** to **5**, the horizontal direction extending from the front side to the rear side of the image forming apparatus M is referred to as an X**1** direction, and the horizontal direction extending from the rear side to the front side is referred to as an X**2** direction. Further, a direction extending from the lower side to the upper side of the image forming apparatus M along the vertical direction (the gravity direction) is referred to as a Z**1** direction, and a direction extending from the upper side to the lower side is referred to as a Z**2** direction. Further, an upper side and a lower side of the image forming apparatus M as viewed from the front side are referred to as an upper surface and a lower surface, respectively. FIG. **2** is a sectional view of the image forming apparatus M as viewed from the non-drive side, and the near side of the paper corresponds to the non-drive side of the image forming apparatus M, the right side of the paper corresponds to the front side of the image forming apparatus M, and the far side of the paper corresponds to the drive side of the image forming apparatus M. The same applies to FIGS. **3** to **5**. Further, in a direction perpendicular to each of the X**1**-X**2** direction and the Z**1**-Z**2** direction, a direction extending from a non-drive side, on which the cartridge **100** does not receive a driving force from the image forming apparatus main body **170**, to a drive side, on which the cartridge **100** receives the driving force from the image forming apparatus main body **170**, is referred to as a Y**1** direction, and a direction opposite to the Y**1** direction is referred to as a Y**2** direction (see FIGS. **6**A and **6**B to **8**).

[0042] In the image forming apparatus main body **170**, first to fourth cartridges **100** (**100**Y, **100**M, **100**C, and **100**K) are arranged in an approximately horizontal direction.

[0043] The first to fourth cartridges **100** (**100**Y, **100**M, **100**C, and **100**K) each have the same electrophotographic process mechanism, and each contain developer (hereinafter, referred to as toner) in different color. The first to fourth cartridges **100** (**100**Y, **100**M, **100**C, and **100**K) receive a rotational driving force transmitted through a drive output portion (main-body-side drum drive couplings **180** and main-body-side developing drive couplings **185** illustrated in FIGS. **3** to **5**) of the image forming apparatus main body **170**. The drive output portion will be described in detail below.

[0044] Further, the first to fourth cartridges **100** (**100**Y, **100**M, **100**C, and **100**K) receive bias voltages (charging bias, developing bias, etc.) supplied from the image forming apparatus main body **170** (not illustrated).

[0045] The first cartridge **100**Y contains yellow (Y) toner and forms a yellow toner image (developer image) on a corresponding photosensitive drum **104**. The second cartridge **100**M contains magenta (M) toner and forms a magenta toner image on a corresponding photosensitive drum **104**. The third cartridge **100**C contains cyan (C) toner and forms a cyan toner image on a corresponding photosensitive drum **104**. The fourth cartridge **100**K contains black (K) toner and forms a black toner image on a corresponding photosensitive drum **104**.

[0046] A laser scanner unit **14** as an exposure unit is provided above the first to fourth cartridges **100** (**100**Y, **100**M, **100**C, and **100**K). The laser scanner unit **14** outputs laser light U toward the surface of the photosensitive drum **104** of each of the first to fourth cartridges **100** (**100**Y, **100**M, **100**C, and **100**K) in accordance with image information supplied by the user.

[0047] An intermediate transfer unit **12** as a transfer member is provided below the first to fourth cartridges **100** (**100**Y, **100**M, **100**C, and **100**K). The intermediate transfer unit **12** includes a driving roller **12***e*, an assist roller **12***c*, and a tension roller **12***b*, and a transfer belt **12***a* having flexibility is stretched around these rollers.

[0048] The photosensitive drum **104** of each of the first to fourth process cartridges **100** (**100**Y, **100**M, **100**C, and **100**K) is positioned above the transfer belt **12***a*, and the outer peripheral surface of each photosensitive drum **104** is in contact with the outer peripheral surface of the transfer belt **12***a* having an annular belt shape. Each contact portion is a primary transfer portion. On the inner side of the transfer belt **12***a*, a primary transfer roller **12***d* is provided to each of the cartridges **100**Y, **100**M, **100**C, and **100**K such that each primary transfer roller **12***d* faces its corresponding photosensitive drum **104**. The driving roller **12***e* contacts a secondary transfer roller **6** via the transfer belt **12***a*. A contact portion between the transfer belt **12***a* and the secondary transfer roller **6** is a secondary transfer portion.

[0049] A paper feed unit **4** is provided below the intermediate transfer unit **12**. The paper feed unit **4** includes a paper feed tray **4***a* that stacks and accommodates the recording medium S and a paper feed roller **4***b*. As illustrated in FIG. **2**, a fixing device **7** and a paper output device **8** are provided in the upper left portion of the image forming apparatus main body **170**. A paper output tray **13** is disposed on the upper surface of the image forming apparatus main body **170**.

Image Forming Operation

[0050] An operation for forming a full-color image will be described with reference to FIGS. **1** and **2**. The photosensitive drum **104** of each of the first to fourth cartridges **100** (**100**Y, **100**M, **100**C, and **100**K) is rotated at a predetermined speed (in the direction of arrow A in FIG. **1**). In addition, the driving roller **12***e* is rotated by a motor (not illustrated) of the image forming apparatus main body **170**, and this rotation moves the transfer belt **12***a* in the direction C in FIG. **2**.

[0051] In each cartridge **100**, a charging roller **105** uniformly charges the surface of the photosensitive drum **104** to predetermined polarity and potential. The laser scanner unit **14** scans and exposes the surface of the photosensitive drum **104** with the laser light U in accordance with an image signal of the corresponding color. As a result, an electrostatic latent image corresponding to the image signal of the corresponding color is formed on the surface of each photosensitive drum **104**. The formed electrostatic latent image is developed with the toner as the developer borne by a developing roller **106**, which is rotated at a predetermined speed.

[0052] By the electrophotographic image forming process operation as described above, a yellow toner image corresponding to a yellow element of a full-color image is formed on the photosensitive drum **104** of the first cartridge **100**Y. Next, the toner image is primarily transferred onto the transfer belt **12***a*. Similarly, a magenta toner image corresponding to a magenta element of the full-color image is formed on the photosensitive drum **104** of the second cartridge **100**M. Next, the magenta toner image is primarily transferred onto the transfer belt **12***a* onto which the yellow toner image has already been transferred in a superimposed manner. Similarly, a cyan toner image corresponding to a cyan element of the full-color image is formed on the photosensitive drum **104** of the third cartridge **100**C. Next, the cyan toner image is primarily transferred in a superimposed manner onto the transfer belt **12***a* where the yellow and magenta toner images have already been transferred. Similarly, a black toner image corresponding to a black element of the full-color image is formed on the photosensitive drum **104** of the fourth cartridge **100**K. Next, the black toner image is primarily transferred in a superimposed manner onto the transfer belt **12***a* where the yellow, magenta, and cyan toner images have already been transferred. In this way, an unfixed full-color toner image using four colors of yellow, magenta, cyan, and black is formed on the transfer belt **12***a*.

[0053] The recording medium S is separated and fed one by one from the paper feed unit **4** at predetermined control timing. The recording medium S is brought into the secondary transfer portion, which is a contact portion between the secondary transfer roller **6** and the transfer belt **12***a*, at predetermined control timing. Thus, during the process of conveying the recording medium S to the secondary transfer portion, the 4-color superimposed toner image on the transfer belt **12***a* is transferred onto the surface of the recording medium S. Next, the toner image transferred onto the recording medium S is fixed by a fixing unit provided in the fixing device **7**, and the recording medium S is discharged to the paper output tray **13** by the paper output device **8**.

Configuration for Attaching and Detaching Cartridge

[0054] A configuration for attaching and detaching the cartridge **100** will be described with reference to FIGS. **3** to

5. FIG. **3** is a sectional view of the image forming apparatus M in a state in which the front door **11** is opened and a tray **171** has been pulled out to the outside of the image forming apparatus main body **170**. FIG. **4** is a sectional view of the image forming apparatus M in a state in which the front door **11** is opened, the tray **171** is positioned outside the image forming apparatus main body **170**, and toner cartridges **101** have been detached from process cartridges **102**. FIG. **5** is a sectional view of the image forming apparatus M in a state in which the front door **11** is opened, the tray **171** is positioned outside the image forming apparatus main body **170**, and the process cartridges **102** have been detached from the tray **171**.

[0055] As illustrated in FIGS. **3** to **5**, the tray **171** is movable in a direction of an arrow X**1** (push-in direction) and a direction of an arrow X**2** (pull-out direction), which are approximately horizontal directions with respect to the image forming apparatus main body **170**. That is, the tray **171** is provided to be capable of being pulled out and pushed in with respect to the image forming apparatus main body **170**, and the tray **171** is configured to be movable in an approximately horizontal direction when the image forming apparatus main body **170** is installed on a horizontal surface. In the present embodiment, a state in which the tray **171** is positioned outside the image forming apparatus main body **170** (the states illustrated in FIGS. **3** to **5**) is referred to as an outside position. Further, a state in which the tray **171** is positioned inside the image forming apparatus main body **170** in a state in which the front door **11** is opened (the state illustrated in FIG. **2**) is referred to as an inside position.

[0056] As illustrated in FIG. **4**, the toner cartridges **101** are detachable from the process cartridges **102**. That is, the cartridges **100** according to the present embodiment are constituted by the process cartridges **102** and the toner cartridges **101**. When the toner runs out and printing cannot be performed, the user can replace the toner cartridge **101** with a new one filled with toner. That is, the toner cartridge **101** whose toner has been consumed is removed from the process cartridge **102**, a new toner cartridge **101** is attached to the process cartridge **102**, the tray **171** is pushed into the image forming apparatus main body **170**, and the front door **11** is closed. In this way, the image formation can be performed again.

[0057] As illustrated in FIG. **5**, the tray **171** includes attaching portions **171**Ya, **171**Ma, **171**Ca and **171**Ka from which the respective process cartridges **102** can be removed in the outside position, and the cartridges **100** of respective colors can be attached on the attaching portions **171**Ya, **171**Ma, **171**Ca and **171**Ka.

[0058] When the photosensitive drum **104**, the developing roller **106**, or the like deteriorates and the printing quality cannot be maintained, the user can replace the process cartridge **102** with a new one. That is, the process cartridge **102** that needs to be replaced is removed from a corresponding one of the attaching portions **171**Ya, **171**Ma, **171**Ca and **171**Ka of the tray **171**, and a new process cartridge **102** is attached. Next, the toner cartridge **101** is attached, the tray **171** is pushed into the image forming apparatus main body **170**, and the front door **11** is closed. In this way, image formation can be performed again.

[0059] In the present embodiment, the intermediate transfer unit **12** can be moved vertically by the front door **11** and a link mechanism (not illustrated). When the front door **11** is opened, the intermediate transfer unit **12** is lowered in the direction of the arrow Z**2**, and the photosensitive drum **104** and the transfer belt **12***a* are spaced from each other. When the front door **11** is closed, the intermediate transfer unit **12** is moved upward in the direction of the arrow Z**1**, and the photosensitive drum **104** and the transfer belt **12***a* are brought into contact with each other. In this way, the tray **171** can move the cartridge **100** into and out of the image forming apparatus main body **170** without a contact between the photosensitive drum **104** and the transfer belt **12***a*.

[0060] As described above, the tray **171** can collectively move the plurality of cartridges **100** into and out of the image forming apparatus main body **170**, and this enables the toner cartridge **101** or the process cartridge **102** to be easily replaced.

Detailed Configuration of Cartridge

[0061] A configuration of the cartridge **100** will be described in detail with reference to FIGS. **1**, **6**A, **6**B, **7**, and **8**. In the present embodiment, the first to fourth cartridges **100** (**100**Y, **100**M, **100**C, and **100**K) each have the same electrophotographic process mechanism, and contain toner T in different colors and different filling levels.

[0062] The cartridge **100** is constituted by a toner cartridge **101** and a process cartridge **102**. The toner cartridge **101** is filled with toner T and supplies the toner T to the process cartridge **102**. The process cartridge **102** stably forms a latent image on the photosensitive drum **104** using the toner T supplied from the toner cartridge **101**, and transfers the latent image to the intermediate transfer unit **12**.

[0063] The toner cartridge **101** includes a toner container **120**, and a toner stirring shaft **121** and toner T are provided in the toner container **120**. A toner stirring sheet **122** is attached to the toner stirring shaft **121**. The toner cartridge **101** receives a driving force transmitted from the process cartridge **102** (not illustrated) and rotates the toner stirring shaft **121** in the direction C to supply the toner T to the process cartridge **102** through a supply opening (not illustrated).

[0064] The process cartridge **102** includes a drum unit **140** and a developing unit **150**.

[0065] The drum unit **140** includes the photosensitive drum **104** as a rotatable image bearing member formed of a photoconductor, the charging roller **105** as a rotatable charging member, and a container **130** as a drum unit frame for holding the photosensitive drum **104** and the charging roller **105**. The charging roller **105** is biased toward the photosensitive drum **104** by a charging spring **105***a*. Further, the surface of the photosensitive drum **104** can be uniformly charged by injecting charges from the image forming apparatus main body **170** to the charging roller **105** through a charging contact point (not illustrated). Further, a drum coupling **160** for transmitting a driving force to the photosensitive drum **104** is provided at one longitudinal end of the photosensitive drum **104**. The drum coupling **160** engages with the main-body-side drum drive coupling **180** (illustrated in FIG. **3**) as a drum drive output portion of the image forming apparatus main body **170**. Thus, the driving force generated by a drive motor (not illustrated) of the image forming apparatus main body **170** is transmitted to the photosensitive drum **104** so that the photosensitive drum **104** is rotated in the direction of the arrow A illustrated in FIG. **1**.

[0066] The developing unit **150** includes the developing roller **106** as a developer bearing member, a supplying roller

107 as a developer supplying member, a developing blade **108** as a regulating member, and a developer container **110** as a developing unit frame. The developer container **110** includes a toner storage chamber **116** that accommodates and holds the toner T as the developer supplied from the toner cartridge **101**, and a developing chamber **117** in which the developing roller **106**, the supplying roller **107**, and the developing blade **108** are disposed. The toner storage chamber **116** is located above the developing chamber **117** and communicates with the developing chamber **117** through a developing opening **110***b*, which is a communication hole penetrating a bottom surface **116***b* of the storage chamber **116** and a top surface **117***b* of the developing chamber **117**.

[0067] A stirring shaft **111** as a rotational shaft is rotatably mounted inside the toner storage chamber **116** of the developer container **110**, and a stirring sheet **109**, which is a sheet member having flexibility (a flexible member), is attached to the stirring shaft **111**. The developing coupling **161** illustrated in FIG. **6**A engages with the main-body-side developing drive coupling **185** (illustrated in FIG. **3**) as a drum drive output portion of the image forming apparatus main body **170**. Thus, the driving force generated by the drive motor (not illustrated) of the image forming apparatus main body **170** can be transmitted to the stirring shaft **111**. The toner T in the toner storage chamber **116** is supplied to the supplying roller **107** through the developing opening **110***b* formed in the developer container **110** by the rotation of the stirring shaft **111** in the direction of the arrow B illustrated in FIG. **1**. The toner T supplied to the supplying roller **107** is supplied to the developing roller **106**. The developing blade **108** regulates the toner T on the developing roller **106** to a uniform thickness. In this way, the toner can be stably supplied to the photosensitive drum **104** with which the developing roller **106** is in contact.

[0068] The drum unit **140** and the developing unit **150** are connected to each other so as to be relatively rotatable about a rotational axis SX parallel to a direction (**Y1-Y2** direction) in which the rotational axes of the photosensitive drum **104** and the developing roller **106** extend. The drum unit **140** and the developing unit **150** are configured to take a contact position (an image forming position) in which the photosensitive drum **104** and the developing roller **106** are brought into contact with each other by receiving a biasing force of a spring (not illustrated). When a force receiving portion **119** provided at a lower end of the developing unit **150** is biased by a force applying portion **190** provided in the image forming apparatus main body **170**, the developing unit **150** moves relative to the drum unit **140** in a rotational direction SD indicated by a dashed arrow. As a result, the relative position between the drum unit **140** and the developing unit **150** transitions from the above-described contact position (the image forming position) to a separate position (a non-image forming position) in which the photosensitive drum **104** and the developing roller **106** are separated from each other. The relative position between the drum unit **140** and the developing unit **150** may be controlled such that the drum unit **140** and the developing unit **150** take the separate position while the image forming operation is not performed, that is, for example, when the cartridge **100** is attached or detached, and take the contact position when the image forming operation is performed. In the present embodiment, the drum unit **140** is fixed to the image forming apparatus main body **170** (the tray **171**), and the developing unit **150** is rotated relative to the drum unit **140** so that the relative position between the drum unit **140** and the developing unit **150** is changed. However, the present invention is not limited to the above-described configuration.

[0069] In the present embodiment, the individual cartridge is constituted by the toner cartridge **101** and the process cartridge **102**. However, the present invention is not limited to the above-described configuration. For example, the toner cartridge **101** and the process cartridge **102** may be integrated into an all-in-one cartridge. Alternatively, the developing unit in which the developing roller and the supplying roller are integrally held in the toner cartridge holding the toner and the drum unit in which the photosensitive drum and the charging roller are held may be configured to be separately provided. Further, the drum unit may have a drum cleaning mechanism.

Method for Detecting Remaining Amount of Toner

[0070] A method for detecting the remaining amount of toner, which is a feature of the present embodiment, will be described in detail with reference to FIG. **1**, and FIGS. **6**A and **6**B to FIGS. **12**A and **12**B. When the user performs printing, the toner is consumed, and the amount of toner in each cartridge **100** decreases. It is desirable to notify the user before the toner is completely used up and an image defect, such as toner not being printed on the printed matter, occurs. Thus, a method for detecting the remaining amount of toner to grasp the amount of toner held in the cartridge **100** will be described.

[0071] As illustrated in FIG. **1**, the stirring shaft **111** is rotatably mounted inside the toner storage chamber **116** of the developer container **110** of the process cartridge **102**. A flexible sheet **112** is fixed to the stirring shaft **111**. The flexible sheet **112** is a sheet member having flexibility, and is fixed to the stirring shaft **111** such that one end of the flexible sheet **112** in a direction crossing the rotational axis line of the stirring shaft **111** is configured to be a free end and an other end is configured to be a fixed end. A magnet **113** as a magnetism generating member is attached at a position away from the other end (the fixed end) fixed to the stirring shaft **111**, more specifically, at a position between the other end exclusive fixed to the stirring shaft **111** and the one end (the free end) inclusive on the opposite side. While the magnet is used as the magnetism generating member in the present embodiment, another material that generates a magnetic force may also be used.

[0072] FIG. **6**A is a perspective view of the cartridge **100**, and FIG. **6**B is a sectional perspective view taken along a cross section K in FIG. **6**A. As illustrated in FIG. **6**B, in the present embodiment, the flexible sheet **112** is provided at a central portion in the longitudinal direction (the **Y1-Y2** direction along the rotational axis line) of the stirring shaft **111**. The stirring sheet **109** for stirring and conveying the toner in the toner storage chamber **116** is provided with a width extending over approximately the entire longitudinal area of the toner storage chamber **116**. It is preferable that the flexible sheet **112** be provided with a minimum width in the longitudinal direction of the toner storage chamber **116** so as not to affect the accuracy of detection of the remaining amount of toner (so as not to greatly change the storage state of toner due to the movement of the flexible sheet **112**). The shape and material of the magnet **113** and the shape, material, and thickness of the flexible sheet **112** can be appropriately selected in accordance with their characteristics.

[0073] A magnetic sensor **114** for detecting magnetism is attached to the container **130** of the process cartridge **102**. The magnetic sensor **114** is not for detecting the magnetism of the magnet **113** provided in the process cartridge **102** to which this magnetic sensor **114** is attached, but for detecting the magnetism of a magnet **113** provided in an other adjacent process cartridge **102** (see FIG. **9**).

[0074] FIG. **7** is a sectional perspective view of the cartridge **100**, taken along a position different from the cross section K in FIG. **6**B, and is a diagram illustrating a configuration of the periphery of the magnetic sensor **114**. FIG. **8** is an enlarged perspective view of the cartridge **100** for describing the configuration of the periphery of the magnetic sensor **114**.

[0075] As illustrated in FIGS. **7** and **8**, the magnetic sensor **114** is held by a holder **114***a*. The holder **114***a* includes guided portions **114***b* that are shaft portions protruding from both ends of the holder **114***a* in a direction (Y**1**-Y**2** direction) along the direction in which the rotational axis line of the stirring shaft **111** extends. The container **130** includes holder guides **114***c* as guiding portions and a sensor spring **115** as biasing members. The holder guides **114***c* and the sensor spring **115** constitute a bias support portion that supports the holder **114***a* such that a biasing force is applied to the holder **114***a*, the biasing force pressing the holder **114***a* against the outer wall surface of an other developer container **110** of an other adjacent cartridge **100**.

[0076] Each of the holder guides **114***c* has a guide groove **114***d* that is hollowed in a direction (Y**1**-Y**2** direction) along the direction in which the rotational axis line of the stirring shaft **111** extends and is extended in a direction in which the holder **114***a* is brought into contact with/separated from an other developer container **110** of an other adjacent cartridge **100**. The guide grooves **114***d* are disposed on both sides of the holder **114***a* in the Y**1**-Y**2** direction, facing each other, and a pair of the guided portions **114***b* provided on both sides of the holder **114***a* in the Y**1**-Y**2** direction are inserted into the respective guide grooves **114***d*. The holder **114***a* is configured to be movable in a direction toward and away from the other developer container **110** (contact/separating direction) by the guided portions **114***b* being guided by the guide grooves **114***d*.

[0077] The sensor spring **115** is attached between the rear surface of the holder **114***a* and the container **130**. The sensor spring **115** applies a biasing force to the rear surface of the holder **114***a*, which is a side opposite to the side on which the magnetic sensor **114** is exposed to the outside, and press the holder **114***a* in a direction toward the other developer container **110**. By receiving such a biasing force, the holder **114***a* is brought into contact with the other developer container **110**, and the magnetic sensor **114** held by the holder **114***a* is brought into a state of facing the outer surface of the other developer container **110**.

[0078] The contact state between the holder **114***a* and an other developer container **110** is automatically formed when each cartridge **100** is attached on the tray **171**, as illustrated in FIGS. **4** and **5**. That is, the other cartridge **100** is configured to be attachable to the tray **171** while the outer wall surface of the other developer container **110** pushes the holder **114***a* away in the separating direction against the biasing force of the sensor spring **115**. As a result, by the attaching of the cartridges **100**, a state in which the holder **114***a* is in contact with the developer container **110** of the adjacent cartridge **100** can be formed.

[0079] A configuration in which the individual magnetic sensor **114** detects the magnetism of the magnet **113** will be described with reference to FIGS. **1** and **9**. FIG. **9** is a schematic sectional view (a sectional view taken along the X**1**-X**2** and Z**1**-Z**2** directions each of which is perpendicular to the rotational axis line of the photosensitive drum **104**, for example) illustrating the tray **171** and the first to fourth cartridges **100**Y, **100**M, **100**C, and **100**K attached on the tray **171**.

[0080] In the present embodiment, among the first to fourth cartridges **100**Y, **100**M, **100**C, and **100**K, the fourth cartridge **100**K includes a magnet **113**K as a magnetism generating member, and does not include a magnetic sensor. The first to third cartridges **100**Y, **100**M, and **100**C are cartridges each of which includes the magnetic sensor **114** for detecting the remaining amount of toner in an other adjacent cartridge **100**. The plurality of cartridges **100** is arranged to be aligned along the X**1**-X**2** direction as a first direction. In the direction (X**1**-X**2** direction) in which the plurality of cartridges **100** is aligned, each magnetic sensor **114** is disposed in the cartridge frame of the cartridge **100** that includes this magnetic sensor **114** such that the magnetic sensor **114** is positioned on a side close to an other cartridge frame of an other adjacent cartridge **100** so as to face the other cartridge frame. That is, in the container **130** of the drum unit **140** and the developer container **110** of the developing unit **150** constituting the cartridge frame, the magnetic sensor **114** is provided in the container **130**, rather than the developer container **110**, and faces the outer surface of the container **130** of an other cartridge **100**. The magnetism of a magnet **113**Y of the first cartridge **100**Y is detected by a magnetic sensor **114**H provided on the tray **171**.

[0081] As illustrated in FIG. **9**, the magnetism of the magnet **113**K of the fourth cartridge **100**K containing black (K) toner is detected by a magnetic sensor **114**C held by the third cartridge **100**C. In this case, a holder **114**aC holding the magnetic sensor **114**C is biased by the sensor spring **115** to be brought into contact with the outer wall surface of the developer container **110**.

[0082] In the relationship of the third cartridge **100**C and the fourth cartridge **100**K to the present invention, the third cartridge **100**C corresponds to a second cartridge of the present invention, and the fourth cartridge **100**K corresponds to a first cartridge of the present invention. The cartridge frame (a developer container **110**C, a container **130**C) of the third cartridge **100**C corresponds to a second frame of the present invention, and the cartridge frame (a developer container **110**K, a container **130**K) of the fourth cartridge **100**K corresponds to a first frame of the present invention. In addition, in the developer container **110**C, the toner storage chamber **116**C corresponds to a second storage chamber of the present invention, the stirring shaft **111** corresponds to a second rotational shaft of the present invention, the flexible sheet **112** corresponds to a second flexible member of the present invention, and the magnet **113**C corresponds to a second magnetism generating member of the present invention. In the developer container **110**K, the toner chamber **116**K corresponds to a first storage chamber of the present invention, the stirring shaft **111** corresponds to a first rotational shaft of the present invention, the flexible sheet **112** corresponds to a first flexible member of the present invention, and the magnet **113**K corresponds to a first magnetism generating member of the present invention. Further, in the third cartridge **100**C, the photosensitive drum **104** corresponds to a second photosensitive drum, the drum unit **140** corresponds to a second drum unit, the developing roller **106** corresponds to a second developing roller, and the developing unit **150** corresponds to a second developing unit. In the fourth cartridge **100**K, the photosensitive drum **104** corresponds to a first photosensitive drum, the drum unit **140** corresponds to a first drum unit, the developing roller **106** corresponds to a first developing roller, and the developing unit **150** corresponds to a first developing unit.

[0083] Similarly, as illustrated in FIG. **9**, the magnetism of the magnet **113**C of the third cartridge **100**C is detected by a magnetic sensor **114**M held by the second cartridge **100**M. At this time, a holder **114**aM holding the magnetic sensor **114**M is biased by the sensor spring **115** to be brought into contact with the outer wall surface of the developer container **110**C.

[0084] In the relationship of the second cartridge **100**M and the third cartridge **100**C to the present invention, the second cartridge **100**M corresponds to the second cartridge of the present invention, and the third cartridge **100**C corresponds to the first cartridge of the present invention. Therefore, the relationship of the respective parts of the second cartridge **100**M and the third cartridge **100**C to the present invention is the same as the relationship of those of the third cartridge **100**C and the fourth cartridge **100**K to the present invention described above.

[0085] In addition, in the relationship of the second to fourth cartridges **100**M, **100**C, and **100**K to the present invention, when the third cartridge **100**C corresponds to the second cartridge of the present invention, the second cartridge **100**M corresponds to a third cartridge of the present invention. In this case, the magnetic sensor **114**C corresponds to a second sensor of the present invention, and the magnetic sensor **114**M corresponds to a third sensor of the present invention.

[0086] Similarly, as illustrated in FIG. **9**, the magnetism of a magnet **113**M of the second cartridge **100**M is detected by a magnetic sensor **114**Y held by the first cartridge **100**Y. In this case, a holder **114**aY holding the magnetic sensor **114**Y is biased by the sensor spring **115** to be brought into contact with the outer wall surface of a developer container **110**M.

[0087] In the relationship of the first cartridge **100**Y and the second cartridge **100**M to the present invention, the first cartridge **100**Y corresponds to the second cartridge of the present invention, and the second cartridge **100**M corresponds to the first cartridge of the present invention. Therefore, the relationship of the respective parts of the first cartridge **100**Y and the second cartridge **100**M to the present invention is the same as the relationship of those of the third cartridge **100**C and the fourth cartridge **100**K to the present invention as described above.

[0088] In addition, in the relationship of the first to third cartridges **100**Y, **100**M, and **100**C to the present invention, when the second cartridge **100**M corresponds to the second cartridge of the present invention, the first cartridge **100**Y corresponds to the third cartridge of the present invention. In this case, the magnetic sensor **114**M corresponds to the second sensor of the present invention, and the magnetic sensor **114**Y corresponds to the third sensor of the present invention.

[0089] Similarly, as illustrated in FIG. **9**, the magnetism of the magnet **113**Y of the first cartridge **100**Y is detected by a magnetic sensor **114**H held by the tray **171**. In this case, a holder **114**aH holding the magnetic sensor **114**H is biased by the sensor spring **115** to be brought into contact with the outer wall surface of a developer container **110**Y.

[0090] In the relationship of the first cartridge **100**Y, the second cartridge **100**M, and the tray **171** to the present invention, the magnetic sensor **114**Y corresponds to the second sensor of the present invention, and the magnetic sensor **114**H corresponds to a fourth sensor of the present invention.

[0091] A method for estimating the amount of toner will be described with reference to FIGS. **10**A to **10**C to FIGS. **11**A to **11**C. While the description will be made for one cartridge **100** here, the amount of toner is detected by the same method in any of the first to fourth cartridges.

[0092] FIGS. **10**A to **10**C are schematic sectional views of the cartridge **100** in a state in which there is nearly full of toner T contained in the developer container **110**. FIGS. **10**A to **10**C illustrate states in each of which the magnet **113** moved by the rotation of the stirring shaft **111** is passing through a detection range R of the magnetic sensor **114** included in the adjacent cartridge **100**. FIG. **10**B is a diagram illustrating a state in which the magnet **113** has reached the position closest to the magnetic sensor **114** disposed in the adjacent cartridge. FIG. **10**A illustrates a state of the rotation of the stirring shaft **111** slightly earlier than the state illustrated in FIG. **10**B. FIG. **10**C illustrates a state of the rotation of the stirring shaft **111** slightly later than the state illustrated in FIG. **10**B.

[0093] FIGS. **11**A to **11**C are schematic sectional views of the cartridge **100** in a state in which there is scarcely any toner T in the developer container **110**. FIGS. **11**A to **11**C illustrate states in each of which the magnet **113** moved by the rotation of the stirring shaft **111** is passing through the detection range R of the magnetic sensor **114** included in the adjacent cartridge **100**. FIG. **11**B is a diagram illustrating a state in which the magnet **113** has reached the position closest to the magnetic sensor **114** disposed in the adjacent cartridge. FIG. **11**A illustrates a state of the rotation of the stirring shaft **111** slightly earlier than the state illustrated in FIG. **11**B. FIG. **11**C illustrates a state of the rotation of the stirring shaft **111** slightly later than the state illustrated in FIG. **11**B.

[0094] The flexible sheet **112** is formed of a sheet member as thin as possible (a PET or PPS sheet having a thickness of 50 μm in the present embodiment) so as not to affect the accuracy of detecting the remaining amount of toner (so as not to greatly change the state of the stored toner by the movement of the flexible sheet **112**). Therefore, as illustrated in FIGS. **10**A to **10**C, in the developer container **110** to which the toner T is abundantly supplied from the toner cartridge **101**, the toner T acts as resistance, and the flexible sheet **112** rotates while being bent when the stirring shaft **111** rotates in the direction B. Therefore, the magnet **113** is separated from the magnetic sensor **114** by a distance of L1 at the position closest to the magnetic sensor **114** (FIG. **10**B).

[0095] On the other hand, as illustrated in FIGS. **11**A to **11**C, when there is scarcely any toner T in the toner cartridge **101**, the toner T is not supplied to the developer container **110**. Since the toner T, which has been serving as resistance, is no longer present in the developer container **110** in which the toner T has been consumed, the amount of bending of the flexible sheet **112** is reduced so that the magnet **113** can be brought closer to the magnetic sensor **114** as illustrated in FIG. **11**B. When this distance between the magnet **113** and the magnetic sensor **114** is represented by L**2**, L**1**>L**2** is obtained. That is, the behavior of the magnet **113** changes in accordance with the amount of toner T in the developer container **110**, and consequently, the duration (hereinafter, referred to as detection time) for which the magnetic sensor **114** detects the magnetism generated by the magnet **113** changes.

[0096] FIG. **12**A is a schematic diagram for describing that the time taken by the magnetic sensor **114** to detect the magnetism of the magnet **113** changes in accordance with the change in the amount of toner. An arrow TrF of a dashed line with one dot illustrated in FIG. **12**A indicates a movement trajectory of the magnet **113** passing through the detection range R of the magnetic sensor **114** while there is nearly full of toner T in the developer container **110**. An arrow TrL of a dashed line with two dots illustrated in FIG. **12**A indicates a movement trajectory of the magnet **113** passing through the detection range R of the magnetic sensor **114** when the toner T in the developer container **110** has been consumed.

[0097] As illustrated in FIG. **12**A, the movement trajectory TrF of the magnet **113** when there is full of toner passes through a position away from the magnetic sensor **114** due to the resistance of the toner. Thus, the magnet **113** stays in the detection range R of the magnetic sensor **114** for a short time. On the other hand, the movement trajectory TrL of the magnet **113** when the toner has been consumed passes through a position closer to the magnetic sensor **114** due to a smaller resistance of the toner. Thus, the magnet **113** stays in the detection range R of the magnetic sensor **114** for a longer time. That is, the smaller the amount of toner is, the closer the magnet **113** can be to the magnetic sensor **114**. Consequently, the detection time increases as the amount of toner decreases.

[0098] FIG. **12**B specifically illustrates the relationship between the amount of toner in the developer container **110** and the detection time of the magnetic sensor **114**. The horizontal axis of the graph in FIG. **12**B represents the amount of toner in the developer container **110**, and the vertical axis represents the detection time. Since the toner T in the developer container **110** is consumed and decreases, the amount of toner actually transitions from the right to the left side of the paper in FIG. **12**B.

[0099] As illustrated in FIG. **12**B, the amount of toner and the detection time are in a proportional relationship until the amount of toner decreases to some extent, and the amount of toner can be estimated (portion D). In a state in which the amount of toner is further reduced from the state of FIGS. **11**A to **11**C (portion E of FIG. **12**B), the remaining amount of toner is extremely small, and the flexible sheet **112** is not bent by the toner T. Therefore, the detection of the remaining amount by the detection configuration of the present embodiment is difficult. Thus, when the remaining amount of toner becomes extremely small, the remaining amount of toner can be estimated by, for example, switching to a remaining amount detection using a method in which the remaining amount of toner is estimated from an image pattern. Although this method has a potential for error accumulation, the impact of such errors would be extremely small since, in the present embodiment, it is only necessary to estimate a small amount of toner, which is nearly used up. Thus, the amount of toner can be detected with high accuracy until the toner filled in the developer container **110** is consumed and used up.

[0100] Further, according to the configuration for detecting the remaining amount of toner in the present embodiment, further space saving and miniaturization of the full-color image forming apparatus can be achieved. In a typical tandem-type image forming apparatus, a predetermined space is provided between adjacent cartridges. However, as illustrated in FIGS. **2**, **9**, etc., the present embodiment utilizes this space for disposing the magnetic sensor **114**.

[0101] As illustrated in FIG. **1**, in the horizontal direction (the X**1**-X**2** direction) approximately along the direction in which the cartridges **100** are aligned, the magnetic sensor **114** is positioned closer to the developer container **110** of the adjacent cartridge **100** than the charging roller **105**. Specifically, in the horizontal direction, the position of the magnetic sensor **114** (a virtual perpendicular line VL**4** passing through the magnetic sensor **114**) is located downstream of the position of the charging roller **105** (a virtual perpendicular line VL**5** passing through the charging roller **105**) in the X**2** direction. In addition, in the vertical direction (the Z**1**-Z**2** direction), the magnetic sensor **114** is positioned higher than the charging roller **105**. Specifically, in the vertical direction, the position of the magnetic sensor **114** (a virtual line HL**4** passing through the magnetic sensor **114**) is higher than the position of the charging roller **105** (a virtual line HL**5** passing through the charging roller **105**). The position of the magnetic sensor **114** corresponds to a dead space formed at the periphery of the charging roller **105** between the adjacent cartridges **100**. According to the present embodiment, the magnetic sensor **114** can be disposed while filling the dead space, and thus, space saving and miniaturization can be achieved.

[0102] The above-described magnetic sensor **114** needs to be disposed such that the magnetic sensor **114** can detect the magnetism of the magnet **113** of the adjacent cartridge **100** without detecting the magnetism of the magnet **113** of the cartridge **100** in which the magnetic sensor **114** is disposed. As illustrated in FIG. **9**, the magnetic sensor **114** is provided so as to be aligned with the toner storage chamber **116** in a direction (the first direction) in which the cartridges **100** are aligned, the direction being approximately along the horizontal direction (the X**1**-X**2** direction). That is, when viewed in the horizontal direction (the X**1**-X**2** direction) approximately along the direction in which the cartridges **100** are aligned, the magnetic sensor **114** is provided at a height overlapping the toner storage chamber **116**. In addition, in the vertical direction (the Z**1**-Z**2** direction), the position of the magnetic sensor **114** (the virtual line HL**4** passing through the magnetic sensor **114**) is higher than the position of the bottom surface **116***b* of the toner storage chamber **116** (the virtual line HL**6** passing through the bottom surface **116***b*). Further, the magnetic sensor **114** is provided at a height overlapping the stirring shaft **111** when viewed in the horizontal direction (the X**1**-X**2** direction). According to the above-described configuration, even if an inexpensive sensor having a relatively narrow detection range R is used as the magnetic sensor **114**, it is possible to provide the arrangement that can increase the magnetism detection time during the rotation of the stirring shaft **111** as the amount of toner contained in the toner storage chamber **116** decreases. As a result, cost reduction can be achieved.

[0103] By detecting the amount of toner in the developer container with the above-described configuration, the remaining amount of toner can be stably detected even with a compact color low-end product. In the present embodiment, the process cartridge in the cartridge system has been described as an example. However, the process cartridge may have a configuration in which the developing device is fixed to the image forming apparatus.

Embodiment 2

[0104] Embodiment 2 of the present invention will be described. In Embodiment 2, configurations and operations different from those in Embodiment 1 will be described. Members having similar configurations and functions as those in Embodiment 1 are denoted by the same reference numerals, and redundant description will be omitted. In Embodiment 2, matters not particularly described are the same as those in Embodiment 1.

[0105] In the present embodiment, an example in which the configuration for detecting the remaining amount of toner according to the present invention is applied to the estimation of the amount of used waste toner will be described.

Detailed Configuration of Cartridge

[0106] A configuration of a cartridge **200** will be described in detail with reference to FIGS. **13** and **14**. In the present embodiment, first to fourth process cartridges **200** (**200**Y, **200**M, **200**C, and **200**K) each have the same electrophotographic process mechanism, and contain toner T in different colors and different filling levels.

[0107] The integrated cartridge **200** includes a rotatable photosensitive drum **104** formed of a photoconductor, a container **230** for holding the photosensitive drum **104**, and a rotatable charging roller **105**. The charging roller **105** is biased toward the photosensitive drum **104** by a charging spring **105***a*. Further, the surface of the photosensitive drum **104** can be uniformly charged by injecting charges from an image forming apparatus main body **170** to the charging roller **105** through a charging contact point (not illustrated). A method for driving the photosensitive drum **104** is the same as that in Embodiment 1, and thus, the description thereof will be omitted.

[0108] The cartridge **200** includes a developer container **210** that holds toner T. A stirring shaft **111** is provided inside the developer container **210**, and a stirring sheet **109** is attached to the stirring shaft **111**. A method for stably supplying the toner T in the developer container **210** to the drum is the same as that in Embodiment 1, and thus, the description thereof will be omitted.

[0109] The cartridge **200** includes a cleaning member **201** fixed to the container **230**. The cleaning member **201** is a member for removing residual toner T**1** remaining on the surface of the photosensitive drum **104** after the primary transfer. The residual toner T**1** removed from the photosensitive drum **104** by the cleaning member **201** can be collected and held in a waste toner portion **240** as a storage chamber formed in the container **230**.

[0110] The cartridge of the present embodiment is an integrated all-in-one cartridge. However, the configuration of the cartridge is not limited thereto. For example, the cartridge may be configured to include a toner cartridge that contains toner and a process cartridge that includes a developing roller, a supplying roller, a drum, a charging roller, and a cleaning member. Alternatively, a developing unit that includes a developing roller, a supplying roller, and toner and a drum unit that includes a photosensitive drum, a charging roller, and a cleaning member may be provided separately.

Method for Detecting Remaining Amount of Toner

[0111] A method for detecting the remaining amount of toner, which is a feature of the present embodiment, will be described in detail with reference to FIGS. **13** to **16**. When the user performs printing, the toner is consumed, and the residual toner T**1** remaining on the drum surface is collected into the waste toner portion **240** of the container **230** by the cleaning member **201**. The capacity of the waste toner portion **240** is limited, and it is necessary to notify the user before the waste toner portion **240** becomes full. In view of this, the present embodiment will describe a configuration in which the user can be notified before the waste toner portion **240** becomes full by detecting the amount of waste toner.

[0112] As illustrated in FIG. **13**, a stirring shaft **211** is rotatably attached to the waste toner portion (waste toner storage chamber) **240** of the container **230**. The stirring shaft **211** can be rotated in an F direction by a drive input (not illustrated). A flexible sheet **212** is fixed to the stirring shaft **211**. The flexible sheet **212** is a sheet member (a flexible member) having flexibility, and is fixed to the stirring shaft **211** such that one end in a direction crossing the rotational axis line of the stirring shaft **211** is configured to be a free end, and an other end is configured to be a fixed end. A magnet **213** as a magnetism generating member is attached to the flexible sheet **212** at a position away from the other end (the fixed end) fixed to the stirring shaft **211**, more specifically, at a position between the other end exclusive fixed to the stirring shaft **211** and the one end (the free end) inclusive on the opposite side. While the magnet is used as the magnetism generating member in the present embodiment, another material that generates a magnetic force may also be used.

[0113] A magnetic sensor **214** for detecting magnetism is attached to the developer container **210**. The magnetic sensor **214** is not for detecting the magnetism of the magnet **213** provided in the cartridge **200** to which this magnetic sensor **214** is attached, but for detecting the magnetism of a magnet **213** provided in an other adjacent cartridge **200** (see FIG. **14**). As in Embodiment 1, in the present embodiment, too, the shape and material of the magnet **213** and the shape, material, and thickness of the flexible sheet **212** can be appropriately selected in accordance with their characteristics.

[0114] As illustrated in FIG. **13**, the magnetic sensor **214** is held by a holder **214***a*. The container **230** is provided with a bias support portion that supports the holder **214***a* such that a biasing force is applied to the holder **214***a* to press the holder **214***a* against the outer wall surface of an other container **230** of an other adjacent cartridge **200**. The specific configuration of the bias support portion including a sensor spring **215** that applies the biasing force is the same as that in Embodiment 1, and thus, the description thereof will be omitted.

[0115] A configuration in which the individual magnetic sensor **214** detects the magnetism of the magnet **213** will be described with reference to FIGS. **13** and **14**. FIG. **14** is a diagram illustrating a state in which the cartridges **200** of the respective colors are attached on the tray **271**. The magnetic sensor **214** is disposed on one side of the developer container **210** in a direction (the X1-X2 direction) in which the plurality of cartridges **200** is aligned, and the magnet **213** is provided in the waste toner unit **240** of the container **230** of each cartridge **200**.

[0116] The magnetism of a magnet **213**Y of the first cartridge **200**Y containing yellow (Y) toner is detected by a magnetic sensor **214**M held by the second cartridge **200**M. In this case, the magnetic sensor **214**M is biased toward the container **230** of the first cartridge **200**Y by the sensor spring **215** illustrated in FIG. **13**.

[0117] Similarly, the magnetism of a magnet **213**M of the second cartridge **200**M is detected by a magnetic sensor **214**C held by the third cartridge **200**C. The magnetic sensor **214**C is biased toward the container **230** of the second cartridge **200**M by the sensor spring **215**, as is the case with the magnetic sensor **214**M.

[0118] The magnetism of a magnet **213**C of the third cartridge **200**C is detected by a magnetic sensor **214**K held by the fourth cartridge **200**K. The magnetic sensor **214**K is biased toward the container **230** of the third cartridge **200**C by the sensor spring **215**, as is the case with the magnetic sensors **214**M and **214**C.

[0119] The magnetism of a magnet **213**K of the fourth cartridge **200**K is detected by a magnetic sensor **214**H held by the tray **271**. The magnetic sensor **214**H is biased toward the container **230** of the fourth cartridge **200**K by the sensor spring **215**, as is the case with the magnetic sensors **214**M, **214**C, and **214**K.

[0120] A method for estimating the amount of toner will be described with reference to FIGS. **15**A, **15**B, and **16**. While the description will be made for one cartridge **200** here, the amount of waste toner in any of the first to fourth cartridges is detected by the same method.

[0121] FIG. **15**A is a sectional view illustrating a state in which there is scarcely any waste toner T**1** in the waste toner portion **240** of the container **230**. In this state, the stirring shaft **211** rotates in the F direction, and the magnet **213** has reached the position closest to the magnetic sensor **214** disposed in the adjacent cartridge. FIG. **15**B is a sectional view illustrating a state in which there is nearly full of waste toner T**1** in the waste toner portion **240** of the container **230**. In this state, the stirring shaft **211** rotates in the F direction, and the magnets **213** has reached the position closest to the magnetic sensor **214** disposed in the adjacent cartridge.

[0122] As illustrated in FIG. **15**A, the cartridge **200** that has been used only for a short time has a small amount of waste toner T**1** stored in the waste toner portion **240** of the container **230**. In this case, the waste toner T**1** does not act as resistance of the flexible sheet **212**, and thus, the flexible sheet **212** is less bent and can be brought closer to the magnetic sensor **214**. FIG. **15**B illustrates a state in which the cartridge **200** has been used for a long time, and the waste toner portion **240** is nearly full of the waste toner T**1**. The flexible sheet **212** is formed of a sheet member as thin as possible (in the present embodiment, a PET or PPS sheet having a thickness of 50 μm). Therefore, as illustrated in FIG. **15**B, the waste toner T**1** acts as resistance, and the flexible sheet **212** rotates while being bent when the stirring shaft **211** rotates in the F direction. Therefore, the magnet **213** and the magnetic sensor **214** are distanced from each other, and a relationship of L**4**>L**3** is obtained.

[0123] As described above, the behavior of the magnet **213** changes in accordance with the amount of the waste toner T**1** in the waste toner unit **240** of the container **230**, and consequently, the detection time for which the magnetic sensor **214** detects the magnetism generated by the magnet **213** changes. That is, as the amount of waste toner decreases, the magnet **213** can be brought closer to the magnetic sensor **214**, and therefore, as the amount of waste toner decreases, the detection time increases.

[0124] FIG. **16** specifically illustrates the relationship between the amount of waste toner in the waste toner portion **240** of the container **230** and the detection time of the magnetic sensor **214**. In the graph of FIG. **16**, the horizontal axis represents the amount of waste toner in the waste toner portion **240**, and the vertical axis represents the detection time. The waste toner T**1** in the waste toner portion **240** of the container **230** increases as the cartridge **200** is used, and therefore, the amount of waste toner actually transitions from the left to the right side of the paper in FIG. **16**.

[0125] As illustrated in FIG. **16**, although the waste toner cannot be detected when the amount of the waste toner is small (portion G), once the amount of the waste toner has increased (portion H), the detection time and the amount of the waste toner have a proportional relationship so that the amount of the waste toner can be estimated.

[0126] By detecting the amount of waste toner in the waste toner unit with the above-described configuration, the user can be notified before the waste toner unit becomes full so that a product with good usability can be provided. Further, the amount of waste toner can be detected in a space-saving manner, and can be stably detected even by a compact color low-end product. As in Embodiment 1, in the present embodiment, too, the process cartridge in the cartridge system has been described as an example. However, the process cartridge may have a configuration in which the developing device is fixed to the image forming apparatus.

[0127] The configuration of each embodiment described above can be combined with the configurations of the others.

[0128] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0129] This application claims the benefit of Japanese Patent Application No. 2024-027478, filed on Feb. 27, 2024, which is hereby incorporated by reference herein in its entirety.

What is claimed is:

1. An image forming apparatus comprising:

an apparatus main body;

a first cartridge that is detachably attachable to the apparatus main body, the first cartridge including a storage chamber that stores developer, a rotational shaft that is provided in the storage chamber so as to be rotatable about a rotational axis line, a flexible member that has one end and an other end in a direction crossing the rotational axis line, the one end being a free end and the other end being a fixed end fixed to the rotational shaft, and a magnetism generating member that is fixed between the fixed end exclusive and the free end inclusive of the flexible member; and

a second cartridge that is detachably attachable to the apparatus main body, the second cartridge including a sensor for detecting magnetism generated by the magnetism generating member.

2. The image forming apparatus according to claim **1**, wherein the first cartridge and the second cartridge are aligned in a first direction, and

wherein the sensor is provided between the first cartridge and the second cartridge in the first direction.

3. The image forming apparatus according to claim **2**, wherein the first cartridge includes a first frame that constitutes the storage chamber, and a developing roller configured to bear developer in the storage chamber,

wherein the second cartridge includes a photosensitive drum and a second frame that rotatably supports the photosensitive drum, and

wherein the sensor is supported by the second frame such that the sensor faces the first frame in the first direction.

4. The image forming apparatus according to claim **3**, wherein, assuming that the photosensitive drum, the storage chamber, and the developing roller are referred to as a second photosensitive drum,

a first storage chamber, and a first developing roller, respectively, the first cartridge includes a first photosensitive drum configured to receive developer supplied from the first developing roller, and

the second cartridge includes a second storage chamber that stores developer and a second developing roller configured to bear developer in the second storage chamber and supply developer to the second photosensitive drum.

5. The image forming apparatus according to claim **2**, wherein the first cartridge includes a photosensitive drum, a cleaning member that removes developer from the photosensitive drum, and a first frame that rotatably supports the photosensitive drum and supports the cleaning member,

wherein the storage chamber is a storage chamber that is constituted by the first frame and that stores developer removed from the photosensitive drum by the cleaning member,

wherein the second cartridge includes a developing roller configured to bear developer and a second frame that rotatably supports the developing roller, and

wherein the sensor is supported by the second frame such that the sensor faces the first frame in the first direction.

6. The image forming apparatus according to claim **3**, wherein, assuming that the photosensitive drum, the storage chamber, and the developing roller are referred to as a first photosensitive drum, a first storage chamber, and a second developing roller, respectively,

the first cartridge includes a first developing roller configured to carry developer and supply developer to the first photosensitive drum, and

the second cartridge includes a second photosensitive drum configured to receive developer supplied from the second developing roller.

7. The image forming apparatus according to claim **3**, wherein the second cartridge includes:

a holder that holds the sensor and that is movable with respect to the second frame; and

a biasing member that applies a biasing force to the holder, the biasing force biasing the holder toward the first frame.

8. The image forming apparatus according to claim **1**, wherein the sensor is disposed at a position where time for detecting the magnetism during rotation of the rotational shaft increases as toner stored in the storage chamber decreases.

9. The image forming apparatus according to claim **1**, wherein, in a case of being viewed in the horizontal direction, the sensor is provided at a height that is higher than a bottom surface of the storage chamber and that overlaps the rotational shaft.

10. The image forming apparatus according to claim **2**, wherein the sensor is provided so as to be aligned with the storage chamber in the first direction.

11. The image forming apparatus according to claim **1**, wherein, assuming that the storage chamber is referred to as a first storage chamber, the rotational shaft is referred to as a first rotational shaft, the rotational axis line is referred to as a first rotational shaft, the flexible member is referred to as a first flexible member, and the magnetism generating member is referred to as a first magnetism generating member,

the second cartridge includes:

a second storage chamber that stores developer;

a second rotational shaft that is provided in the second storage chamber so as to be rotatable about a second rotational axis line;

a second flexible member that has one end and an other end in a direction crossing the second rotational axis line, the one end being a free end and the other end being a fixed end fixed to the rotational shaft; and

a second magnetism generating member that is fixed between the fixed end and the free end of the second flexible member, and

wherein the sensor is disposed at a position where the sensor does not detect magnetism generated by the second magnetism generating member.

12. The image forming apparatus according to claim **11**, wherein the first cartridge includes a first frame including the first storage chamber,

wherein the second cartridge includes a second frame including the second chamber, and

wherein, in an alignment direction of the first cartridge and the second cartridge,

the sensor is provided in the second frame on a side close to the first frame, and

the second storage chamber is provided in the second frame on a side away from the first frame.

13. The image forming apparatus according to claim **12**, wherein the second cartridge includes a photosensitive drum and a charging roller for charging the photosensitive drum,

wherein, in the alignment direction,

the photosensitive drum and the charging roller are provided in the second frame on a side close to the first frame, and

the sensor is provided at a position higher than the charging roller and closer to the first frame than the photosensitive drum and the charging roller.

14. The image forming apparatus according to claim **12**, wherein the second cartridge includes a photosensitive drum, a developing roller that bears developer for developing an electrostatic latent image formed on the photosensitive drum, a third storage chamber that stores developer to be supplied to the developing roller, and a cleaning member for removing developer from the photosensitive drum,

wherein the second storage chamber is a storage chamber that stores developer removed from the photosensitive drum by the cleaning member, and

wherein, in the alignment direction,

the third storage chamber and the developing roller are provided in the second frame on a side close to the first frame and

the sensor is provided at a position closer to the first frame than the third storage chamber and the developing roller.

15. The image forming apparatus according to claim **11**, wherein the first cartridge includes:

a first drum unit that includes a first photosensitive drum; and

a first developing unit that includes a first developing roller which bears developer for developing an electrostatic latent image formed on the first photosensitive drum, the first storage chamber which stores developer to be supplied to the first developing roller, the first rotational shaft, the first flexible member, and the first magnetism generating member,

wherein the second cartridge includes:

a second drum unit that includes a second photosensitive drum; and

a second developing unit that includes a second developing roller which bears developer for developing an electrostatic latent image formed on the second photosensitive drum, the second storage chamber which stores developer to be supplied to the second developing roller, the second rotational shaft, the second flexible member, and the second magnetism generating member,

wherein the first drum unit, the first developing unit, the second drum unit, and the second developing unit are aligned in this order, and

wherein the sensor is provided in the second drum unit.

16. The image forming apparatus according to claim **15**, wherein the second developing unit is configured to be movable with respect to the second drum unit such that the second developing roller can take a contact position and a separate position with respect to the second photosensitive drum.

17. The image forming apparatus according to claim **11**, wherein the first cartridge includes:

a first drum unit that includes a first photosensitive drum, a first cleaning member for removing developer from the first photosensitive drum, the first storage chamber which stores developer removed from the first photosensitive drum by the first cleaning member, the first rotational shaft, the first flexible member, and the first magnetism generating member; and

a first developing unit that includes a first developing roller which bears developer for developing an electrostatic latent image formed on the first photosensitive drum,

wherein the second cartridge includes:

a second drum unit that includes a second photosensitive drum, a second cleaning member for removing developer from the second photosensitive drum, the second storage chamber which stores developer removed from the second photosensitive drum by the second cleaning member, the second rotational shaft, the second flexible member, and the second magnetism generating member; and

a second developing unit that includes a second developing roller which bears developer for developing an electrostatic latent image formed on the second photosensitive drum,

wherein the first developing unit, the first drum unit, the second developing unit, and the second drum unit are aligned in this order, and

wherein the sensor is provided in the second developing unit.

18. The image forming apparatus according to claim **11**, further comprising a third cartridge that is detachably attachable to the apparatus main body,

wherein the third cartridge is attached to the apparatus main body so as to be adjacent to the second cartridge on a side opposite to the first cartridge, and

wherein, assuming that the sensor is referred to as a second sensor, the third cartridge includes a third sensor for detecting magnetism generated by the second magnetism generating member.

19. The image forming apparatus according to claim **11**, wherein the apparatus main body includes a tray that supports the first cartridge and the second cartridge, and

wherein, assuming that the sensor is referred to as a second sensor, the tray includes a fourth sensor for detecting magnetism generated by the second magnetism generating member.

20. A cartridge comprising:

a storage chamber that stores developer;

a rotational shaft that is provided in the storage chamber so as to be rotatable about a rotational axis line;

a flexible member that has one end and an other end in a direction crossing the rotational axis line, the one end being a free end and the other end being a fixed end fixed to the rotational shaft; and

a magnetism generating member that is fixed between the fixed end exclusive and the free end inclusive of the flexible member,

wherein the cartridge is one of a plurality of cartridges detachably attachable to an apparatus main body of an image forming apparatus, and

wherein the cartridge includes a sensor for detecting magnetism generated by an other magnetism generating member included in an other cartridge among the plurality of cartridges attached to the apparatus main body.

21. The cartridge according to claim **20**,

wherein the sensor does not detect magnetism generated by the magnetism generating member included in the cartridge.

22. The cartridge according to claim **20**,

wherein the cartridge is aligned with the other cartridge in a first direction, and

wherein the sensor is provided between the cartridge and the other cartridge in the first direction.

23. The cartridge according to claim **20**,

the cartridge comprising:

a photosensitive drum; and

a frame that rotatably supports the photosensitive drum, wherein the sensor is supported by the frame such that the sensor faces an other frame included in the other cartridge in the first direction.

24. The cartridge according to claim **23**,
the cartridge comprising:
a holder that holds the sensor and that is movable with respect to the other frame; and
a biasing member that applies a biasing force to the holder, the biasing force biasing the holder toward the other frame.

25. The cartridge according to claim **20**,
wherein the sensor is disposed at a position where time for detecting magnetism generated by the other magnetism generating member during rotation of an other rotational shaft included in the other cartridge increases as toner stored in an other storage chamber included in the other cartridge decreases.

26. The cartridge according to claim **20**,
wherein, in a case of being viewed in the horizontal direction, the sensor is provided at a height that is higher than a bottom surface of an other storage chamber included in the other cartridge and that overlaps an other rotational shaft included in the other cartridge.

27. The cartridge according to claim **20**,
the cartridge comprising:
a drum unit that includes a photosensitive drum; and
a developing unit that includes a developing roller which bears developer for developing an electrostatic latent image formed on the photosensitive drum, the storage chamber which stores developer to be supplied to the developing roller, the rotational shaft, the flexible member, and the magnetism generating member;
wherein the sensor is provided in the drum unit.

28. The cartridge according to claim **27**,
wherein the drum unit includes a charging roller for charging the photosensitive drum, and
wherein the sensor is provided at a position higher than the charging roller and closer to the other cartridge than the photosensitive drum and the charging roller.

29. The cartridge according to claim **20**,
the cartridge comprising:
a drum unit that includes a photosensitive drum, a cleaning member for removing developer from the photosensitive drum, the storage chamber which stores developer removed from the photosensitive drum by the cleaning member, the rotational shaft, the flexible member, and the magnetism generating member; and
a developing unit that includes a developing roller which bears developer for developing an electrostatic latent image formed on the photosensitive drum,
wherein the sensor is provided in the developing unit.

30. The cartridge according to claim **29**,
wherein the developing unit includes a second storage chamber that stores developer to be supplied to the developing roller, and
wherein the sensor is provided at a position closer to the other cartridge than the second storage chamber and the developing unit.

31. The cartridge according to claim **20**,
wherein magnetism generated by the magnetism generating member is detected by another sensor included in still another cartridge attached to the apparatus main body.

32. The cartridge according to claim **20**,
wherein the cartridge is supported by a tray included in the apparatus main body, and
wherein magnetism generated by the magnetism generating member is detected by an other sensor provided in the tray.

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